Master's Thesis

Multilevel Representation Learning for Generative Diffusion Transformers

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Declaration

I hereby declare, that I am the sole author and composer of my thesis and that no other sources or learning aids, other than those listed, have been used. Furthermore, I declare that I have acknowledged the work of others by providing detailed references of said work.
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Abstract

State-of-the-art driving world models operate in a next-frame generation paradigm, by predicting future observations based on the representation of past input. Diffusion Transformers (DiTs) have emerged as a powerful approach within this paradigm, which uses VAE-based tokenizer representations for high perceptual fidelity. However, effective prediction in autonomous driving requires representations that are not only visually accurate but also semantically meaningful and geometrically grounded. In this work, we systematically study methods to enrich these representations by making changes at the input, output, architecture and objective levels using various masking schemes and distillation techniques. Finally, we propose a framework combining JEPA-inspired objectives within a Masked Autoencoder (MAE) to capture organized, multilevel scene structures.

Our analysis reveals that auxiliary supervision from semantic priors provides a more organized latent space, which is well-suited for DiT-based image generation. While masking and JEPA-inspired MAE objectives yield perceptually stable and semantically strong latent tokens, they result in suboptimal image generation quality. We conclude that for DiT architectures, the spatial and structural organization of the latent space is as critical as the semantic richness of individual tokens. Our work demonstrates various tokenizer designs for multilevel representation learning that preserves both semantic and perceptual qualities for the tokens in continuous latent space. Finally, we evaluate and discuss the impact of these representations on downstream image generation task. The implementation and model checkpoints are available on GitHub.

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Contents

List of Figures

List of Tables

1. Introduction

1.1. Motivation

Diffusion models have emerged as a powerful class of generative models by learning to reverse a stochastic noise corruption process. As the field moves beyond static image generation toward robust world modeling, the architectural focus has shifted toward the use of Diffusion Transformers (DiTs) [1] for sequence generation. In practice, the tokens used in these models are typically derived from pretrained autoencoding frameworks such as VAEs [2], which are primarily optimized for visual fidelity. However, a fundamental tension arises in this design: while traditional tokenizers are effective at capturing pixel-level texture, tasks such as world modeling require representations that also encode high-level semantic and geometric information alongside perceptual detail. This motivates our investigation into multilevel representation learning, an approach that enriches the latent space by integrating semantic identity, geometric structure, and perceptual fidelity. By moving beyond purely pixel-based objectives, we aim to determine whether such a hybrid latent space can provide the grounded foundation necessary for generative agents to understand and operate within complex environments.

The shift toward multilevel representations is particularly critical for the development of world models in autonomous driving. Unlike standard generative models, a world model must function as a simulator that understands scene structure and uses the latent space to predict future environmental states, such as in next-frame prediction. Motivated by this need for grounded simulation, we evaluate our multilevel representation learning approach through Diffusion Transformer-based image generation. If a model can accurately represent a novel environment, this provides evidence that the latent space is not only perceptually robust but also sufficiently semantically and geometrically rich to support reliable long-term world simulation.

The limitations of reconstruction-centric tokenizers highlight the need for representation learning approaches that bridge the gap between semantic expressiveness, geometric stability, and visual fidelity. Recent advances in self-supervised learning (SSL) offer a promising direction, enabling models to extract meaningful features without explicit labels. Frameworks such as Masked Autoencoders [3], knowledge distillation [4] and modern predictive architectures like JEPA [5] have demonstrated a stronger ability to encode high-level abstractions compared to standard pixel-level objectives. Despite their success in discriminative tasks such as classification and detection, these SSL methods are typically optimized for representation quality in isolation. As a result, they are not explicitly designed to preserve the delicate balance between semantic richness and perceptual fidelity. In this work, we investigate the integration of these self-supervised objectives into the tokenization process to study the trade-off between semantic richness and local detail. We further evaluate how this balance influences the performance of Diffusion Transformers in generating high-quality, novel images.

By moving beyond simple autoencoding, we incorporate diverse masking strategies, auxiliary supervision through distillation-based objectives, and temporal compression to maintain a balance between perceptual fidelity and semantic richness in continuous latent tokens. Our systematic analysis focuses on four core research dimensions:

RQ1. Input Configuration and Context: How do variations in masking strategies and temporal context influence the balance between semantic richness and perceptual fidelity in the continuous latent tokens?

RQ2. Teacher-Student Dynamics: To what extent do pretrained teacher models and architectural variations demonstrate the effectiveness of knowledge distillation for multilevel representation learning?

RQ3. Temporal Compression: How does temporal compression affect tokenizer quality?

RQ4. Hybrid Supervision: Can a JEPA-inspired, EMA-stabilized self-distillation framework, combined with MAE, improve latent token representation? Furthermore, how does the effectiveness of this supervision vary across continuous latent space, discrete token space, and decoder feature space?

To address these research questions, we conduct a series of controlled experiments. We evaluate the resulting latent spaces in terms of reconstruction fidelity, semantic segmentation quality, and geometric consistency. Finally, we assess whether these latent tokens can be effectively leveraged for high-quality image generation.

1.2. Contributions

To advance the development of semantically grounded, geometrically consistent, and perceptually stable latent spaces for Diffusion Transformer-based image generation, the primary contributions and insights of this work are summarized as follows:

- **Masking analysis:** We perform both spatial and temporal masking across a range of masking ratios and strategies. Our results show that high random-range masking ratios help maintain perceptual, semantic and geometric quality in the latent space. Additionally, spatial masking yields more balanced latent representations compared to temporal masking.
- **Auxiliary supervision using knowledge distillation:** We introduce an auxiliary supervision loss that aligns student latent tokens with features extracted from pretrained teacher models. Our experiments demonstrate that

incorporating semantic and geometric priors from pretrained models such as DINO, Depth, and RAFT enables a strong balance between perceptual fidelity, semantic richness and geometric consistency in the learned tokens.

- **Temporal compression:** Using a fixed masking strategy and temporal context, we evaluate the ability of tokens derived from a single frame to reconstruct both target and context frames. While this approach yields marginal improvements in semantic quality, more substantial gains are observed when combined with strong semantic supervision.
- **EMA-supervison:** Inspired by Joint Embedding Predictive Architectures (JEPA), we investigate the effect of supervision from an EMA-stabilized teacher model on masked inputs. We analyze this supervision separately in the continuous latent space, discrete token space, and decoder feature space. Our findings suggest that indirect supervision in the decoder feature space improves the semantic and geometric quality of latent tokens compared to a standard VQ-VAE baseline.
- **Image generation evaluation:** We identify optimal tokenizer configurations that achieve a robust balance between semantic richness, geometric consistency and perceptual fidelity. Our results show that auxiliary supervision through distillation induces a more structured latent space that is well suited for Diffusion Transformer-based image generation, yielding high-quality results (see Figure 1, Figure 2). However, we also observe that models achieving the best balance between semantic and perceptual quality, particularly those using masked settings, can lead to suboptimal generative fidelity.

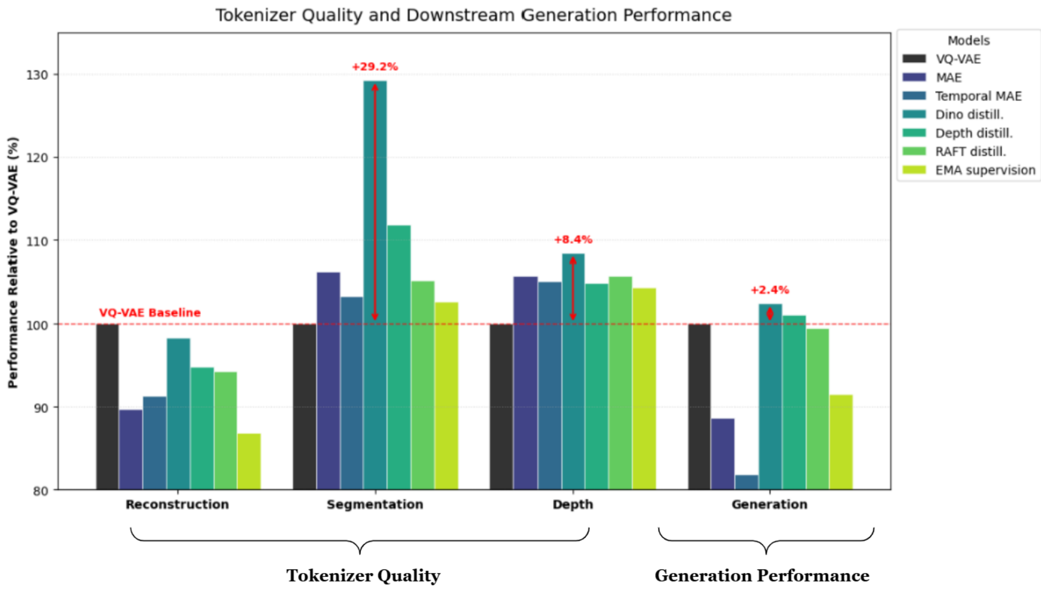


Figure 1.: Multilevel representation learning and image generation. A comparative study of performance across different tokenizer configurations. All proposed tokenizers demonstrate improvements in semantic and geometric quality over the baseline, with auxiliary supervision-based models (DINO, Depth) achieving superior Diffusion Transformer (DiT)-based generation quality.

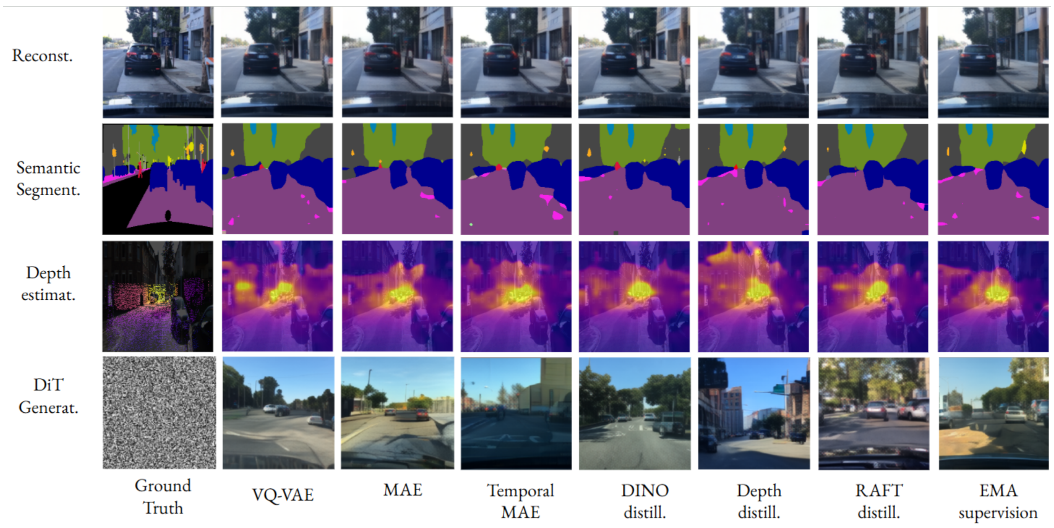


Figure 2.: Visualizations for multilevel representation learning based tokenizers. A visualization comparing perceptual (reconstruction), semantic (segmentation), geometric (depth), and generative performance across different tokenizer configurations.

1.3. Overview of Thesis

This thesis is structured into six chapters that guide the reader from theoretical foundations to empirical analysis and future directions. **Chapter 2** establishes the technical background, covering an overview of world modeling, flow matching in Diffusion Transformers, and the core principles of multilevel representation learning. **Chapter 3** reviews related work, tracing the recent convergence of reconstruction-centric methods and modern self-supervised strategies such as masking, auxiliary supervision (distillation), and prediction. **Chapter 4** details the proposed methodology, including the tokenizer architecture, dataset specifications, evaluation protocols, and training setup. In **Chapter 5**, we present a systematic analysis of the results, discussing performance trade-offs across the different research dimensions.

Finally, **Chapter 6** summarizes the findings, reflects on the limitations of the current framework, and proposes directions for future research in structured latent space design for Diffusion Transformer-based image generation.

2. Background

2.1. World Modeling

World modeling, as illustrated in Figure 3, involves learning a predictive model of an environment that captures both latent states and their dynamics. The model operates by encoding the current state S_t into a latent representation and using a dynamics model to predict the next state S_{t+1}, conditioned on an action a_t. This formulation enables the system to simulate future trajectories and evaluate the consequences of actions within a compressed latent space. The idea originates from cognitive science, where systems are assumed to maintain internal models of their environment [6]. In modern artificial intelligence, this concept is formalized through World Models, which typically consist of an encoder, a latent dynamics model, and a controller.

World models are widely used in robotics, computer vision, and reinforcement learning to enable sample-efficient learning, multi-step planning, and improved generalization to unseen environments. They allow agents to reason at a higher level, anticipate rare events, and optimize actions without extensive real-world interaction. In autonomous driving, for example, world models predict future trajectories of vehicles and pedestrians through next-frame or multi-step prediction, supporting safe

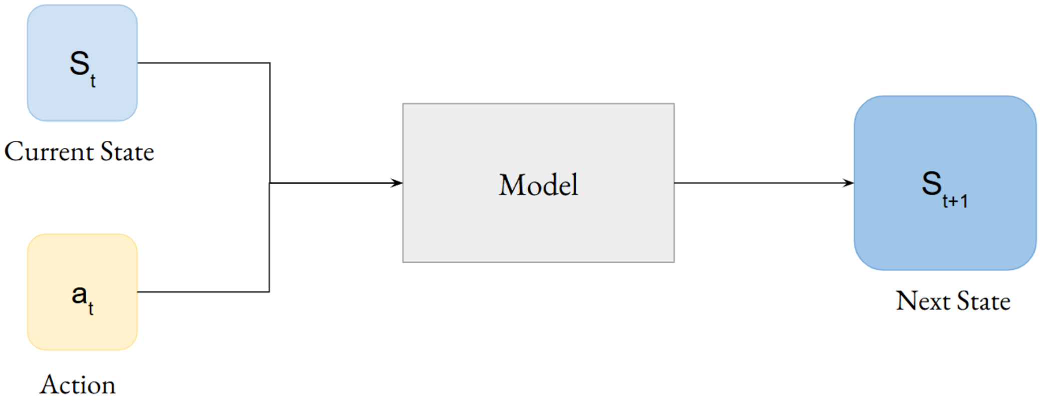


Figure 3.: World modeling in artificial intelligence. The model functions as a predictor that estimates the next state S_{t+1} by processing the current state S_t and action a_t. The definition of action and next state is task-dependent. For instance, in next-frame prediction for autonomous driving, the state is typically a visual observation, while the action represents the vehicle's control inputs or inferred temporal momentum.

and proactive decision-making. More generally, world modeling enables agents to anticipate, plan, and act effectively in complex and dynamic environments.

2.2. Flow Matching

This section describes a generative framework for evaluating learned latent representations. A Diffusion Transformer (DiT) operating in latent space is used for image generation, where the model learns an iterative denoising process that maps Gaussian noise to structured images. In this formulation, the training objective is to predict the velocity of the transformation between noisy and clean samples, introduced by Lipman [7]. Generation is therefore obtained by integrating a learned vector field, rather than performing discrete denoising steps.

Given an input image x_0, we first encode it into a latent representation using a pretrained tokenizer: $z_0 = E(x_0)$, where $E(.)$ denotes the encoder network. A random diffusion timestep $t \sim \mathcal{U}(0, 1)$ is sampled, and noise is added using a forward process:

$$z_t = \alpha(t)z_0 + \sigma(t)\epsilon, \qquad \epsilon \sim \mathcal{N}(0,1) \tag{2.1}$$

Here, $\alpha(t)$ controls the signal strength, while $\sigma(t)$ controls the noise scale at timestep t. This produces noisy latent z_t, which serves as the input to the Diffusion Transformer (DiT) model.

The DiT takes noisy latent z_t as input, patchifies the latent into a sequence of tokens, and processes them using transformer blocks with positional embeddings. The model is trained to predict the velocity field, defined as:

$$\hat{v}_\theta = f_\theta(z_t, t) \tag{2.2}$$

where $\hat{v}_\theta$ represents the model's predicted direction of transformation from noisy latent back toward the data distribution.

The corresponding target velocity represents the true flow direction induced by the forward diffusion process. The model is optimized using a mean-squared error regression objective, which encourages the predicted velocity field to match the true flow at all noise levels:

$$L = E_{z_0,t,\epsilon}[||\hat{v}_\theta - v||^2] \tag{2.3}$$

During inference, generation begins from Gaussian noise: $z_T \sim N(0,1)$ and the model iteratively denoises the latent using the Euler-style updates:

$$z_{t-\Delta t} = z_t + \hat{v}_\theta(z_t)\Delta t \tag{2.4}$$

After the final step, the resulting latent is decoded back into image space: $\hat{x} = D(z_0)$, where $D(.)$ denotes the decoder network.

This generative framework allows us to evaluate whether the learned latent space preserves sufficient semantic and geometric structure to support high-quality image generation.

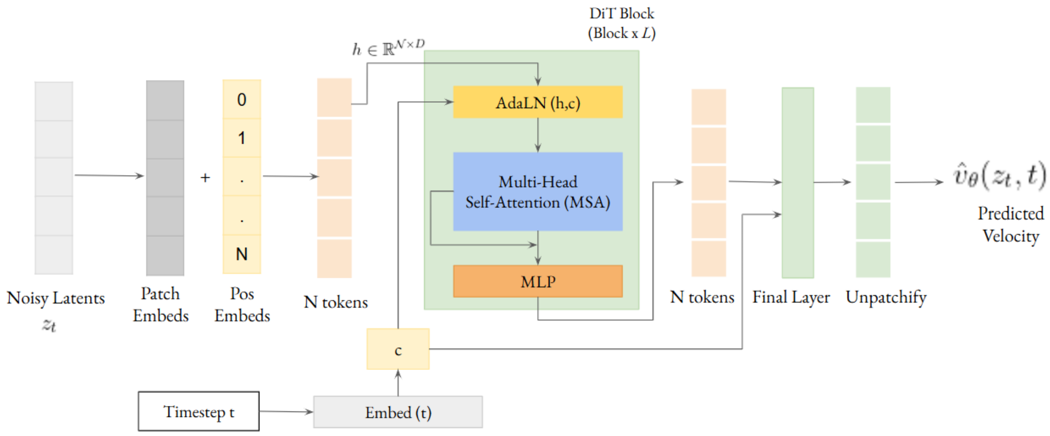


Figure 4.: Diffusion Transformer (DiT) architecture: A noisy latent representation z_t is patchified into tokens and combined with 2D positional embeddings. The diffusion timestep t is embedded into a conditioning vector c, which modulates all Transformer blocks via Adaptive Layer Normalization (AdaLN). The token sequence is processed through stacked Transformer layers with self-attention and MLPs to capture global interactions. Finally, a timestep-conditioned prediction head maps the tokens back to the spatial domain via unpatchifying, producing the predicted velocity field $\hat{v}_\theta(z_t, t)$.

2.3. Diffusion Transformers

As explained in section 2.2, flow matching is trained to predict the velocity field f_θ. This velocity predictor $f_\theta(.)$ is implemented using a Diffusion Transformer (DiT), which operates directly in the latent space. The model (Figure 4) takes as input a noisy latent representation z_t (Equation 2.1) and the corresponding diffusion timestep t, and predicts the velocity field (Equation 2.2) required to reverse the corruption process.

The noisy latent $z_t \in \mathbb{R}^{C\times H\times W}$ is first converted into a sequence of $\mathcal{N}$ tokens using a patch embedding operation, followed by adding 2D sinusoidal positional embeddings. This transforms the spatial latent into a sequence of $\mathcal{N}$ tokens of dimension D, i.e., $z \in \mathbb{R}^{\mathcal{N}\times D}$. The diffusion timestep $t \in [0, 1]$ is embedded into a vector representation $c = Embed(t)$. This embedding encodes the noise level of the input and is used to condition all Transformer layers.

Instead of directly concatenating the timestep embedding to the input tokens, DiT incorporates conditioning through Adaptive Layer Normalization (AdaLN). For a given hidden representation h (i.e., the token representation z_l at layer l), the normalization is defined as :

$$\text{AdaLN}(h, c) = \gamma(c).\text{LN}(h) + \beta(c) \tag{2.5}$$

where, LN(.) is a standard layer normalization, $\gamma(c)$ and $\beta(c)$ are learnable functions of the timestep embedding c. Here, $h \in \mathbb{R}^{\mathcal{N} \times D}$ represents the sequence of token embeddings obtained after patch embedding and positional encoding, and subsequently refined through the Transformer layers. This mechanism allows the model to dynamically adjust its feature scaling and shifting based on the noise level, enabling different denoising behavior across timesteps.

The token sequence is then processed through a stack of Transformer blocks consisting of multi-head self-attention (MSA) and feedforward networks (MLPs). For a given layer l, the hidden representation is updated as:

$$z_{l+1} = z_l + \text{MSA}(\text{AdaLN}(z_l, c)) + \text{MLP}(\text{AdaLN}(z_l, c)) \tag{2.6}$$

Through self-attention, the model captures global dependencies across the latent representation. After the final Transformer layer, the tokens are mapped back to the spatial domain using a prediction head followed by an unpatchifying operation.

The output has the same dimensionality as the input latent and represents the predicted velocity field. The predicted output $\hat{v}_\theta(z_t, t)$ corresponds to the velocity field and is trained using the objective to match the predicted velocity field with the true flow (Equation 2.3). By conditioning on the timestep through AdaLN and leveraging global attention, the Diffusion Transformer provides a flexible and expressive architecture for learning continuous denoising dynamics in latent space.

2.4. Representation Learning

This section discusses representation learning and how the quality of learned latent tokens influences generative performance, particularly in terms of fidelity and semantic consistency. It further introduces multilevel representation learning, in which tokens encode both low-level details and high-level semantics, and explains how this structure improves image generation quality.

Representation learning is the process of learning compact, informative encodings of raw, high-dimensional data x that captures the underlying structure and semantics needed for downstream tasks. Formally, a model defines a mapping:

$$f_\phi : x \to z \tag{2.7}$$

where z is a latent representation capturing the essential factors of variation in the data.

In the context of Vision Transformers (ViTs), Figure 5, the input image $x \in \mathbb{R}^{H\times W\times C}$ is first patchified into non-overlapping patches and projected into a set of patch embeddings. Let $x_p \in \mathbb{R}^{N\times D}$ denote the sequence of N embedded patches, where D is the embedding dimension. This sequence along with the positional embeddings is processed through a stack of Transformer encoder layers yielding latent representation z. Each Transformer layer refines the representation through self-attention mechanisms, allowing the model to capture global dependencies between image regions.

The final representation z serves as a global descriptor of the image and can be used for downstream tasks such as classification, object detection, and semantic segmentation.

Quality of learned latent tokens in representation learning:

From a representation learning perspective, the quality and structure of learned

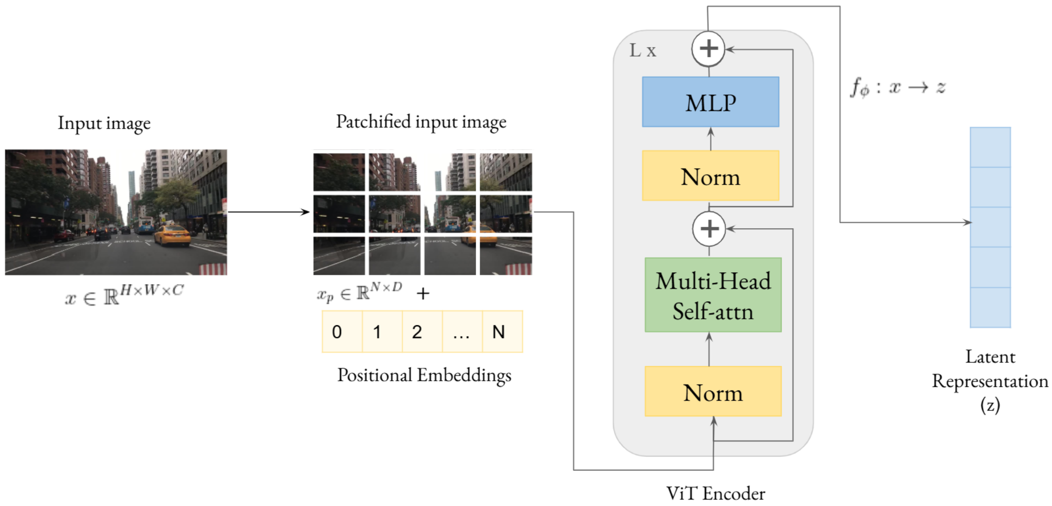


Figure 5.: Representation learning using ViT encoder. An input image x is patchified into non-overlapping patches x_p, learned positional embeddings are added to it and x_p is passed through a multilayer ($L\times$) ViT encoder to generate the latent representation z.

tokens z depend on the training objective used to optimize the model parameters ϕ. Specifically, the mapping f_ϕ is learned by minimizing an objective function $\mathcal{L}$, which implicitly determines what information is preserved in the latent space. As a result, the learned representation z encodes features that are most relevant to the target objective while potentially discarding other information.

Example: When trained with a pixel-level reconstruction loss, the objective is to reconstruct the input image from its latent representation. Under this setting, the learned tokens z are encouraged to retain fine-grained, low-level information such as texture, color, and spatial details, since accurate reconstruction requires preserving most of the input signal. This illustrates that the properties of learned tokens are directly governed by the choice of training objective.

Multilevel representation learning:

multilevel representation learning involves capturing information at multiple levels of abstraction. On the one hand, representations preserve pixel-level details such as edges, textures, and local spatial structures that are closely tied to the raw input.

On the other hand, they encode semantic information such as object identity and scene context.

The balance between these levels largely depends on the training objective. Objectives such as pixel reconstruction encourage the preservation of fine-grained visual details, whereas discriminative objectives promote more abstract and semantically meaningful features. As a result, the learned representation reflects a trade-off between low-level fidelity and high-level semantics. The quality of learned representations plays a critical role in the performance of generative models. Since these latent tokens are used for generation, their ability to preserve meaningful structure directly impacts the fidelity and realism of the generated images. Poorly structured representations can lead to loss of detail or weak semantic consistency in the output, while expressive representations enable more accurate and coherent generation.

2.4.1. Self-supervised Learning

In generative modeling, the quality of these learned representations directly determines the fidelity and coherence of generated data, as generation typically proceeds by sampling z and producing $x \sim p_\theta(x|z)$ [8]. Poorly structured latent spaces often lead to incoherent or semantically weak outputs, motivating the need for multilevel representations that capture both low-level details and high-level structure. In practice, such representations are increasingly learned through self-supervised learning objectives, such as masking, reconstruction, or predictive tasks, to encourage models to capture meaningful spatial and temporal dependencies without requiring dense annotations. These objectives implicitly shape the latent space, promoting representations that are both information-rich and semantically structured, forming the basis for effective generation and prediction in complex environments.

In order to learn such structured and hierarchical representations, a range of self-supervised learning (SSL) approaches can be broadly categorized into complementary families based on how they shape the learned representations:

- **Masked modeling:** Masked modeling learns representations by predicting missing parts of an input from its visible context. In vision, this is done by masking a subset of image patches and training the model to reconstruct them, which encourages both local feature understanding and global structural reasoning. Formally, an image x is divided into tokens $\{x_1, x_2, ..., x_N\}$, and a subset is masked $M \subset \{1, ..., N\}$. The model is trained to minimize the reconstruction loss over the masked tokens:

$$\mathcal{L}_{mask} = \sum_{i \in M} l(x_i, \hat{x}_i) \tag{2.8}$$

where $\hat{x}_i$ is the predicted token, which can be reconstructed in pixel space, feature space, or discrete token space. This forces the model to learn dependencies across patches without requiring labels. Masked modeling leverages structural relationships in data to learn rich hierarchical representations and has become a key technique in Vision Transformers for capturing both low-level details and high-level semantics.

- **Distillation-based methods:** Distillation-based methods aim to learn representations by transferring knowledge from a teacher model to a student model, encouraging consistency in their latent representations. Unlike reconstruction-based methods, which rely on recovering input data or features via pixel reconstruction, distillation directly aligns representations with the target (teacher) model. Formally, given a student representation $z_{student} = f_\phi(x)$ and a teacher representation $z_{teacher} = f_\phi{}'(x)$, the objective minimizes a similarity loss:

$$L_{distill} = L(z_{student}, z_{teacher}) \tag{2.9}$$

In distillation methods, the teacher can be a strong pre-trained model that provides supervisory signals to guide the student toward meaningful representations (knowledge distillation), or it could be a copy of a student encoder, typically updated using an exponential moving average (EMA) of the student parameters (self-distillation), producing a slowly evolving target that stabilizes training and prevents representation collapse.

Overall, these self-supervised learning strategies provide complementary mechanisms for learning semantic and multilevel visual representations.

2.5. Representation Evaluation

Representation learning quality is typically evaluated through two complementary perspectives: perceptual fidelity and semantic utility. Perceptual metrics assess how well a model preserves visual realism in generated or reconstructed images, while semantic evaluation focuses on how effectively the learned representations support downstream tasks. Together, these criteria help determine whether a representation captures both low-level visual structure and high-level semantic information.

1. **Fréchet Inception Distance:**

   The quality of generated and reconstructed images can be measured using perceptual metrics, Fréchet Inception Distance for generation and reconstruction (gFID and rFID). Both measure the distance between feature distributions of two sets of images using embeddings extracted from a pre-trained Inception network [9].

   Formally, given two distributions with feature statistics $(\mu_1, \sum_1)$ and $(\mu_2, \sum_2)$, the FID score is defined as:

   $$\text{FID} = \|\mu_1 - \mu_2\|_2^2 + \text{Tr}\Big(\Sigma_1 + \Sigma_2 - 2(\Sigma_1\Sigma_2)^{1/2}\Big) \tag{2.10}$$

   where, $||\mu_1 - \mu_2||_2^2$ measures the distance between the average feature vectors of two sets μ_1 and μ_2, while the covariance term compares the diversity of distribution.

   Generation FID (gFID) measures the distance between real images and generated samples, evaluating the quality and diversity of generative model. Reconstruction FID (rFID) measures the distance between original images and their reconstructions, assessing how well the model preserves perceptual and semantic information. Lower FID values indicate closer alignment between distributions, reflecting higher image quality.

2. **Linear Probing:**

   Linear probing evaluates the semantic quality of learned representations by measuring how well they support downstream tasks using a simple linear classifier. The idea is that a high-quality representation should encode features that are linearly separable.

   Formally, given frozen encoder features $z = f_\phi(x)$, a linear classifier W is trained on top of the latent representation z to predict task-specific labels y. The linear probe minimizes a standard classification loss, such as cross-entropy:

   $$\mathcal{L}_{\text{linear}} = \frac{1}{N} \sum_{i=1}^{N} \ell(Wz_i, y_i) \tag{2.11}$$

   where N is the number of labeled samples.

   Strong performance under a linear probe indicates that the representation captures high-level semantic features that are useful for downstream tasks like classification, segmentation or depth estimation. Unlike reconstruction metrics, linear probing focuses on the transferability and discriminative power of the latent space, rather than low-level fidelity.

3. Related Work

Recent works on diffusion-based world models explore different ways to improve structured representation learning beyond reconstruction-based latents. GAIA-2 [10] extends earlier world modeling approaches like GAIA1 [11] by achieving better semantic consistency by combining richer conditioning signals with improved structured scene representations. DriveDreamer [12] demonstrates that incorporating structured scene conditioning improves controllability and consistency in generated driving scenes. Epona [13] further improves temporal modeling by introducing an autoregressive diffusion formulation that better captures long-horizon dependencies in dynamic scenes. WorldRFT [14] and MacTok [15] suggest that continuous, robust tokenization can prevent representation collapse. In addition, Dreamer-style world models [16] demonstrate that learning compact latent dynamics for prediction can yield more task-relevant representations than purely pixel-level reconstruction. Finally, geometry-aware diffusion approaches [17],[18],[19],[20] show that incorporating explicit spatial or geometric constraints can improve geometric consistency and enhance the structural coherence of generated scenes. MAETok [21] further argues that the variational constraints of VAEs [2] are unnecessary, and that a discriminative latent space, learned through self-supervision, empowers faster and higher-quality generation. In our work, we build on this trend by investigating how multilevel objectives can bridge the gap between high-fidelity reconstruction and semantically rich tokenization.

3.1. Self-supervised Representation Learning

3.1.1. Masked Modeling

Self-supervised learning (SSL) with masked modeling has become a strong approach for learning visual representations. In images, methods such as Masked Autoencoders (MAE) [3] and SimMIM [22] show that masking large portions of the input (around 75%) encourages models to capture more global structure and semantic information. This idea is also adopted in modern vision backbones such as ConvNeXt V2 [23] which integrates masked image modeling into stronger architecture designs. MaskFeat [24] introduced the use of HOG features as reconstruction targets to capture local orientation. MAGE [25] unifies masked encoding with image synthesis. In videos, where temporal redundancy is high, methods such as VideoMAE [26], [27] and MAE-ST [28] use even higher masking ratios (up to 90%) to learn spatiotemporal representations capturing appearance and temporal dynamics. MAETok [21] demonstrated that predicting a combination of low-level (pixels) and high-level (DINOv2, CLIP) targets creates the most effective tokenizer for diffusion. We build on this idea by enforcing reconstruction from highly incomplete inputs and extending it to both spatial and temporal settings in our framework.

3.1.2. Self-Distillation and Feature Alignment

Teacher-student frameworks like DINO [29] and BYOL [30] learn strong visual representations using a momentum-updated teacher network that provides stable training targets. In contrast, MoCo [31], follows a contrastive learning formulation with a momentum encoder that maintains consistent negative and positive representations rather than explicit distillation. DINOv2 [32] scales this framework through large-scale training and improved optimization strategies, leading to more generalizable representations. Another line of work uses pretrained teacher models for supervision, where methods like DeiT [33] introduces distillation tokens to transfer knowledge from pretrained teacher models, while BEiT [34] and iBOT [35] extend supervision to patch-level or token-level representations using masked prediction and self-distillation objectives. Multi-teacher approaches such as DMT [36] further improve the quality of representations by combining multiple supervision signals. We extend this paradigm by using task-specific teacher models, to provide complementary supervision signals for learning improved representations.

3.1.3. Hierarchical Supervision

Building on hierarchical and predictive representation learning, recent methods explore richer supervision signals beyond final-layer or reconstruction-based objectives. Discrete-JEPA [37] predicts discrete semantic tokens instead of continuous embeddings, while DualToken [38] separates latent space into pixel-level tokens for fine details and semantic tokens for high-level structure. Hierarchical teacher supervision leverages the intermediate hierarchy of pretrained models for distillation, where Data2Vec [39] uses aggregated multi-layer teacher features as prediction targets, while Bootleg [40] supervises students using multiple intermediate teacher layers to capture different abstraction levels. Methods such as SDSSL [41] and DeSD [42] further extend this idea by applying deep supervision across multiple student layers to improve feature learning throughout the network. In this work, we investigate the impact of supervision spaces along with masking and EMA stabilized teacher on the representation quality of the latent tokens.

Overall, recent advances in representation learning increasingly move beyond purely reconstruction-based objectives toward structured supervision, hierarchical distillation, and latent-space prediction. These strategies raise important questions regarding the trade-off between local detail preservation and semantic abstraction. Our work investigates this by systematically comparing diverse self-supervised learning strategies and evaluating their effectiveness for downstream image generation.

4. Approach

In this chapter, we present our methodology for learning latent representations that capture both high-level semantic structure and low-level perceptual detail. Our approach focuses on the design of a multilevel tokenizer and the formulation of its training objectives. Beyond the architectural framework, we describe the evaluation protocols used to assess the effectiveness of these representations, as well as the autonomous driving datasets employed. Finally, we detail the training setup and hyperparameter settings used in our implementation.

4.1. VQ-VAE Baseline

As a baseline, we adopt the Vector Quantized Variational Autoencoder (VQ-VAE) framework, shown in Figure 6 , which learns image representations through a pixel-level reconstruction objective. The architecture, follows the original VQ-VAE formulation [43]. An encoder maps high-dimensional images to a continuous latent space that is subsequently quantized into discrete codebook entries and decoded back to the pixel space.

This choice is motivated by the role of the tokenizer as the primary data provider for Diffusion Transformers (DiT). DiTs operate on sequences of latent tokens rather than raw pixels. The use of a discrete codebook is a common design choice in generative modeling [43], as it helps regularize the latent space and is widely adopted in image generation frameworks. In this setting, it enables the compression of complex

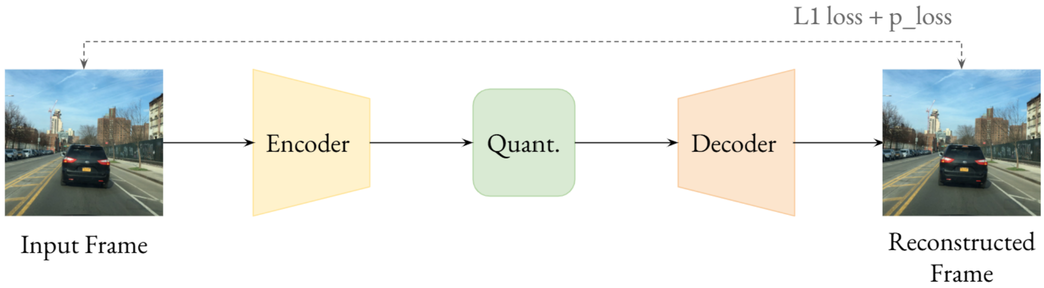


Figure 6.: Architecture: Baseline VQ-VAE model. The VQ-VAE [43] model compresses input images into a discrete latent space via a vector-quantized bottleneck, utilizing a combination of $\mathcal{L}_1$ and perceptual losses to ensure high-fidelity reconstruction.

autonomous driving scenes into compact token sequences while preserving the essential structural information required for stable diffusion. Overall, this setup provides a robust and controlled baseline for evaluating how different training objectives and architectural modifications affect representation quality and downstream evaluation metrics.

4.2. Experimental Factors

In the following, we focus on the design and analysis of the tokenizer, which constitutes Stage 1 of our pipeline. All experimental variations introduced in this work are applied at this stage. To evaluate the quality of the learned representations, the resulting continuous latents are subsequently used in Stage 2 for linear probing and generative modeling. This second stage serves as an evaluation framework to assess how well the learned tokens support downstream tasks.

4.2.1. Input Configurations

This subsection studies the effect of different input configurations on the learned representations. While the VQ-VAE architecture largely remains unchanged (Figure 6),

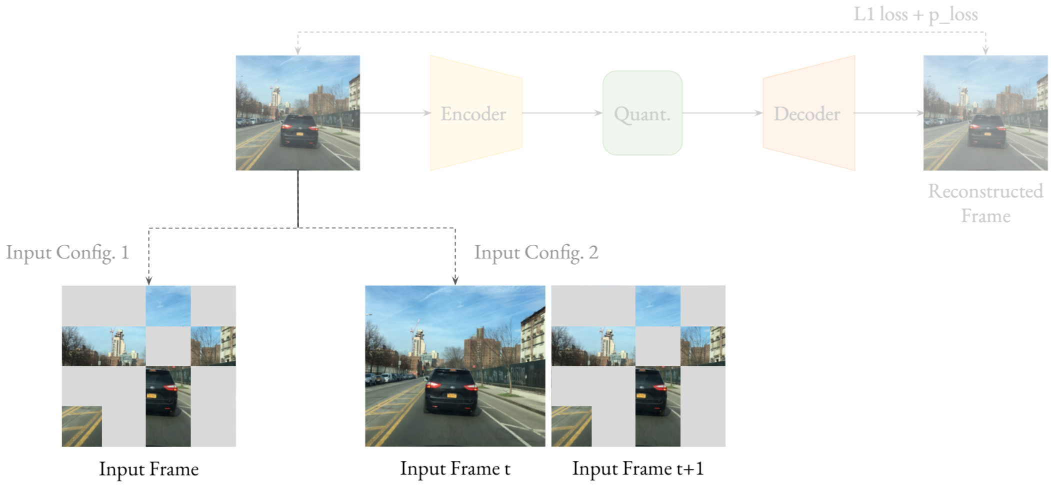


Figure 7.: Architecture: Modified input configurations for VQ-VAE baseline. The VQ-VAE model reconstructs the input frame. To enhance latent representation quality, we investigate two primary input configurations. **Config 1** employs masked image modeling, where portions of the input frame are masked during training, **Config 2** incorporates temporal context (frame t) as input to reconstruct the target (frame t+1), encouraging semantically richer representations.

we modify the input provided to the encoder to analyze its impact on the latent space.

Input Configuration - 1: Masked Image Modeling

Masked Image Modeling (MIM) is a self-supervised representation learning approach in which parts of an input image are masked, and the model is trained to reconstruct the missing regions. This encourages the learning of context-aware representations that capture both local details and global structure.

In our setup, we mask a fraction of the input patches by replacing them with learnable tokens and train the tokenizer to reconstruct the input frame.
Formally, an input frame x is divided into a set of non-overlapping patches $x = \{x_1, x_2, ..., x_N\}$, which are embedded into patch representations. A subset of these embeddings is replaced with mask tokens, resulting in x' , which is then processed by

a Vision Transformer (ViT) encoder to produce latent tokens $z = f_\phi(x')$ as output. These tokens are vector-quantized and passed to a decoder to reconstruct the image $\hat{x} = g_\theta(z)$.

The model is trained using a combination of L1 and perceptual loss:

$$\mathcal{L} = \lambda_1 \sum_i ||x_i - \hat{x_i}||_1 + \lambda_2 \sum_i ||\phi(x_i) - \phi(\hat{x_i})||_2^2$$

where, $\phi(.)$ denotes the Learned Perceptual Image Patch Similarity (LPIPS) metric, computed using a pretrained VGG16 backbone with learned linear calibration layers, following [44] and λ_1, λ_2 balance the contribution of the two loss terms.
In our experiments, we vary the masking ratio to study its effect on representation quality. We further consider two masking strategies: 'Fixed masking', where a constant mask ratio is maintained during training, and 'Range masking', where the ratio is randomly sampled within a predefined range.

Overall, this setup allows the tokenizer to learn representations that capture both local and global context. Based on the performance, we select the best masking strategy and mask ratio to evaluate it under image generation.

Input Configuration - 2: Temporal Context-based Image Modeling

Temporal Context-based Image Modeling extends VQ-VAE and Masked Image Modeling (MIM) setups to sequential data by incorporating information from adjacent video frames. Instead of relying on a single frame, the model leverages temporal context to better infer missing regions and capture motion-related information.

In this setup (Figure 7), the tokenizer is provided with two temporally adjacent input frames $\{x_t, x_{t+1}\}$, where parts of the input frames are masked during training. These two frames are jointly processed using self-attention, allowing information to be exchanged across temporal boundaries. The tokenizer is trained to reconstruct the target frames x_{t+1}, while frame x_t serves as an additional context in this setting. The training is performed using reconstruction objective with the same L1 and perceptual losses as used in Input Configuration 1.

We start with a simple baseline setting where both the frames are fully visible (full context setting) and experiment with three masking strategies, as shown in Figure 8:

1. **Single-frame Masking**: Target frame x_{t+1} is masked, by replacing masked tokens with learnable tokens, while context frame x_t is fully visible.
2. **Asymmetric Masking:** Target frame x_{t+1} is masked, by replacing masked tokens with learnable tokens, while context frame x_t drops the masked patches at the same positions as in target frame.
3. **Heavy Range Masking:** Target frame x_{t+1} and context frame x_t are heavily masked, where tokens from context frame are dropped, while those in target frame are learnable.

These strategies control how information is used across the two frames by varying the balance between spatial and temporal cues. This allows us to study how temporal context influences the representation quality. We identify the best setting based on the representation quality and evaluation it for image generation task.

4.2.2. Auxiliary Supervision through Knowledge Distillation

In this section, we extend the baseline tokenizer design by adding an auxiliary distillation loss, while keeping the original VQ-VAE baseline unchanged. The goal is to examine whether feature-level supervision from a pretrained teacher can improve the learned representations without altering the core pipeline.

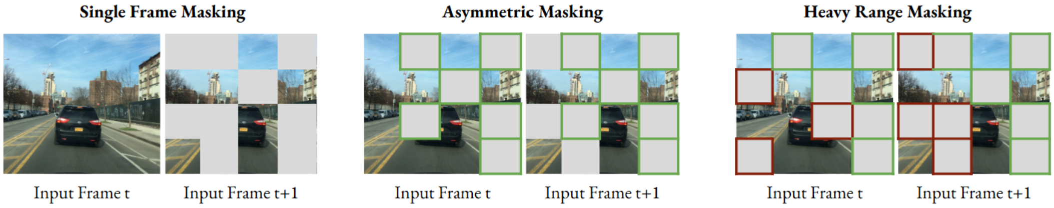


Figure 8.: Masking Strategies : Temporal context-based image modeling. We evaluate three masking strategies : (a) **Single-frame masking:** Masking only frame t+1 with learnable tokens, (b) **Asymmetric masking:** Using heavy token-based masking for frame t+1 and light patch-dropping for frame t at identical locations (marked green), (c) **Heavy range masking:** Employing aggressive, independent random masking across both frames to maximize the reconstruction challenge - masking at identical (marked green) and nonidentical (marked red) locations.

To incorporate auxiliary supervision, we pass an input image through the ViT encoder (Figure 9). The encoded features are duplicated and processed through a separate quantization pathway for indirect supervision. The resulting tokens are projected into a 768-dimensional embedding space and aligned with teacher features using a cosine similarity loss. In parallel, the original VQ-VAE branch remains unchanged and reconstructs the input image. This dual-branch design decouples the learned features from the VQ-VAE codebook and prevents interference with the reconstruction objective. The overall training objective is given by:

$$L = \lambda_{rec} L_{rec} + \lambda_p L_{percept} + \lambda_d L_{distill}$$

The loss weights (λ) are fixed to 1.0 in our experiments, as no training instability or divergence is observed. We monitor individual loss components throughout the training to verify stable optimization.

We use multiple pretrained teacher models, including DINOv2 (semantic features), Depth Anything V2 (depth-aware features), and RAFT (motion cues). Additionally, we incorporate a heavily masked pretrained MAE model from our MIM setup (subsection 4.2.1 - Input Configuration 1) to study self-knowledge distillation within

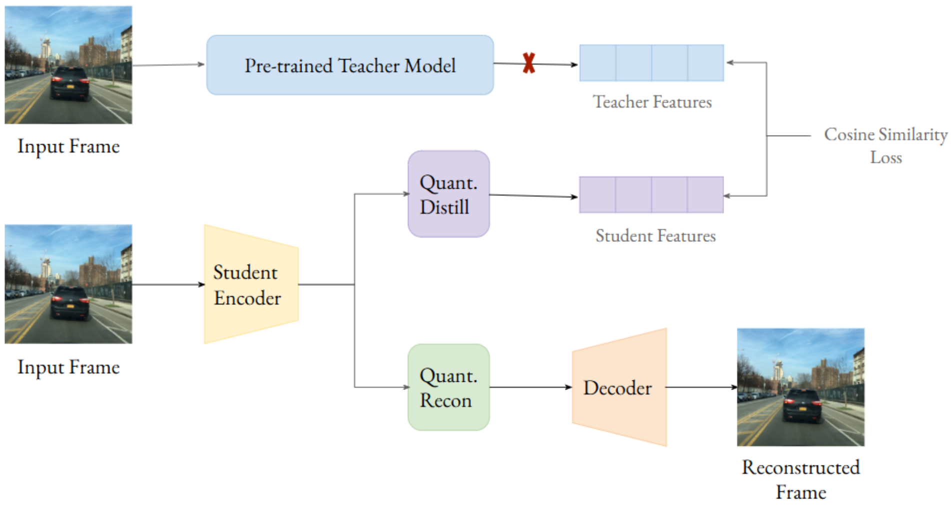


Figure 9.: Architecture: Auxiliary supervision through knowledge distillation. We introduce an auxiliary distillation branch that aligns quantized tokens with embeddings from a frozen teacher network. A cosine similarity loss is computed in a 768-dimensional feature space, while gradients are prevented from flowing into the teacher model. This encourages the model to learn semantically grounded representations.

the same framework. Each teacher is applied independently to analyze the effect of different supervision signals.

Overall, the unified tokenizer is optimized jointly using reconstruction and distillation losses, enabling it to learn richer and more structured representations.

4.2.3. Cross-frame Temporal Reconstruction

Building on Input Configuration 2 (subsection 4.2.1), where temporal context is used to improve representation learning, we further explore whether this idea can be strengthened by explicitly enforcing temporal consistency in the reconstruction objective. In particular, we investigate whether reconstructing both the context and target frames leads to improved latent representation quality.

In this setting, the model (Figure 10) takes two consecutive input frames $\{x_t, x_{t+1}\}$ and processes them jointly using a ViT encoder. Although both frames are pro-

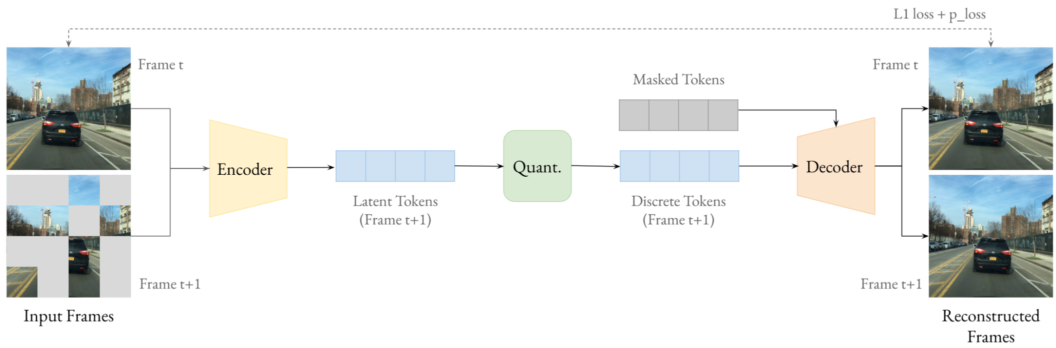


Figure 10.: Architecture: Cross-frame temporal reconstruction. Two consecutive frames are processed jointly using a ViT encoder. Further only the target frame embeddings t+1 undergo vector quantization. Mask tokens are introduced in the decoder to enable reconstruction of both the input frames t and t+1.

vided as input, only the representation corresponding to frame x_{t+1} is retained as the encoder output. These latents are then vector-quantized and passed to the decoder. Since the tokens only from the frame x_{t+1} are processed, learnable mask tokens are added in the decoder to represent missing positions corresponding to frame x_t. In addition, spatial conditioning is introduced on the masked tokens via learned spatial position embeddings and temporal conditioning via frame embeddings. This enables the model to distinguish between frames while preserving spatial structure. The decoder reconstructs both frames from a single set of quantized representations, allowing temporal information to be shared within a unified latent space. The 'Single-frame masking' strategy from Input Configuration 2 (subsection 4.2.1) is adopted, as masking has consistently shown better representation quality. This also enables a direct comparison of the proposed architectural modification against temporal context-based image modeling. To further enhance this setup, we introduce auxiliary supervision using a DINO-pretrained teacher, allowing us to assess how additional semantic guidance improves performance.

Overall, we investigate how enforcing cross-frame temporal reconstruction influences the quality of learned latent representations and their effectiveness in downstream evaluation tasks.

4.2.4. EMA-stabilized Teacher Supervision for MAE

In this section, we explore a JEPA-inspired approach for latent representation learning. JEPA [45] captures high-level semantic features by predicting target latent representations from partially observed (masked) inputs, forcing the model to rely on contextual and structural information rather than low-level details. A slowly updated target encoder is used to provide stable supervision. While this formulation produces strong semantic representations, our objective is to design a tokenizer that is not only semantically rich, but also balances pixel level details.

To obtain semantic representations, we integrate similar supervision idea into our autoencoder-based framework. In this setting, a masked student encoder learns to align with features from an EMA-stabilized teacher encoder, while the decoder preserves the reconstruction objective using VQ-VAE setup. This enables joint learning of semantic features and pixel-level details. A vector quantizer introduces a structural separation in the representation space. The continuous encoder features retain rich and expressive information, which is well-suited for semantic alignment, whereas the quantized representations are constrained to a discrete codebook optimized for reconstruction and generative consistency. This distinction raises a key design question: at which stage should supervision be applied to best balance semantic abstraction and reconstruction fidelity?

To address this, we investigate how to balance global semantic structure and local reconstruction details. We further explore, whether direct supervision (applied in the continuous latent space) or indirect supervision (applied in the post quantization stage) leads to better improvements over a baseline VQ-VAE model. The overall training objective is given by:

$$L = \lambda_{rec} L_{rec} + \lambda_p L_{percept} + \lambda_{ema} L_{ema}$$

We experiment with different loss weights (λ) in our experiments. We monitor individual loss components throughout the training to verify stable optimization.

In the following, we explain how we approach and design each direct and indirect supervision method for multilevel representation learning:

1. **Direct Supervision: Supervision in continuous latent space**

   In this variant, we apply feature-level supervision in the continuous latent space. As shown in Figure 11, a masked input image is passed through an encoder. The encoded latents are further passed through a lightweight predictor to align student and teacher features using a cosine similarity loss, while the decoder receives the features via another branch for reconstruction.

   We investigate two teacher signal choices: (i) features from the first attention block, capturing local, low-level information, and (ii) features from the full attention block, capturing more global and semantically rich representations. This allows us to study how the depth and abstraction level of teacher features affect supervision quality. In both cases, the student encoder uses the same masking strategy, and the predictor maps student features to the corresponding teacher space.

   Overall, this setup isolates the effect of continuous-space supervision while keeping the VQ-VAE pipeline intact, and separately evaluates the impact of shallow versus full-depth teacher representations.

2. **Indirect Supervision: Supervision in the post-quantization feature space**

   a) Discrete Latent Space Supervision:

   In this setting (Figure 12), we apply feature-level supervision in the discrete latent space, immediately after vector quantization. This setup allows us to study how enforcing alignment on quantized tokens affects representation learning compared to continuous-space supervision. An input image is first passed through the student encoder, and the resulting features are vector-quantized

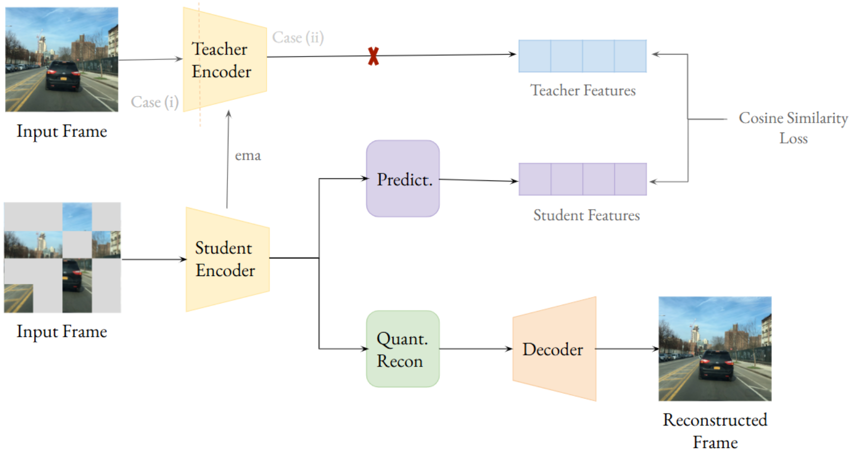


Figure 11.: Architecture: EMA-stabilized teacher supervision in continuous latent space. A teacher–student framework is employed, where student features derived from masked inputs are aligned with an EMA-stabilized teacher using a cosine similarity loss. In parallel, a reconstruction branch reconstructs the input image within the VQ-VAE framework. To analyze the impact of feature hierarchy, two supervision signals are considered: (i) Signal from only the Block 1 of the teacher encoder, (ii) Signal from the entire teacher encoder block.

and projected back into the latent embedding space. A separate branch then aligns these quantized tokens with teacher features using a cosine similarity loss, while these tokens are further passed through the decoder for reconstruction.

Overall, this setup analyzes how discrete-space supervision influences representation quality.

b) Decoded Feature Space Supervision: In this variant (Figure 13), we apply feature-level supervision in the decoded feature space of the VQ-VAE. Instead of operating on continuous latents or quantized representations, the alignment is performed during decoding, where features are closer to the final reconstruction space (specifically the features are considered before the upsampling block). The VQ-VAE architecture and reconstruction objective remain unchanged. We introduce supervision between the decoded student features

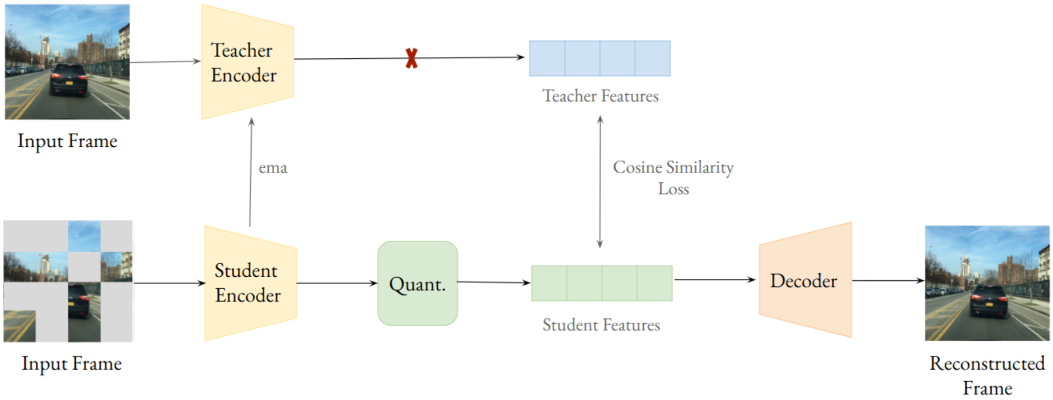


Figure 12.: Architecture: EMA-stabilized teacher supervision in discrete latent space. An EMA-stabilized teacher–student framework is employed, where supervision is applied in the discrete latent space. A cosine similarity loss aligns student and teacher representations, with gradients propagated only through the student encoder. In parallel, the decoder reconstructs the input using the VQ-VAE framework, enabling learning of semantically guided discrete representations.

and the corresponding teacher representations, encouraging consistency at a higher semantic level. This setup allows us to study whether aligning features in a more reconstruction-aligned space provides a better balance between representation quality and reconstruction fidelity compared to continuous latent and discrete space supervision.

4.3. Evaluation Protocols

To comprehensively evaluate the quality of the learned latent representations, we adopt multiple complementary protocols, with image generation as the primary objective. We analyze the representation quality of the tokens by balancing perceptual fidelity and semantic quality. Finally, we evaluate how these learned representations work in the DiT block for image generation.

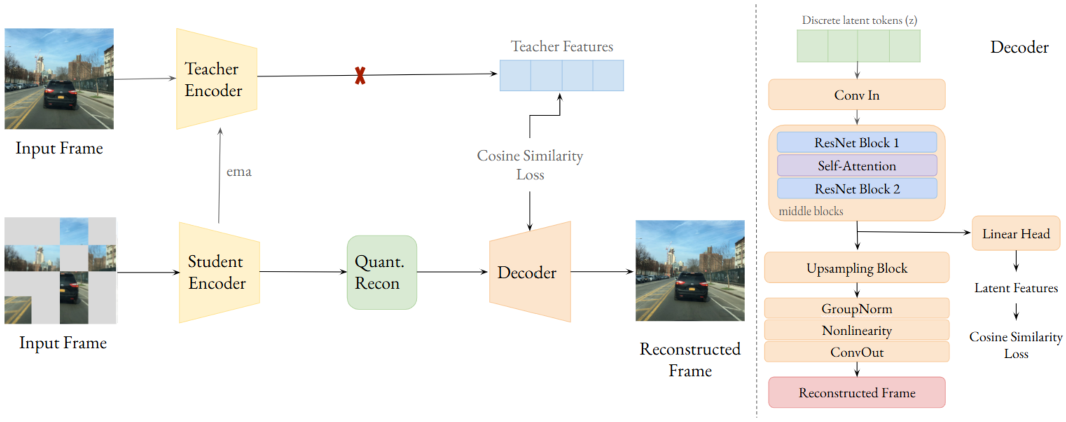


Figure 13.: Architecture: EMA-stabilized teacher supervision at decoded feature space. An EMA-stabilized teacher–student framework is employed, where supervision is applied in the decoded feature space. Student features from masked inputs are first quantized and passed through the decoder. Features extracted before the decoder upsampling stage are aligned with those from an EMA-stabilized teacher encoder using a cosine similarity loss, with gradients propagated only through the student encoder. The aligned features are then upsampled and reconstructed, enabling supervision at the decoded feature level.

4.3.1. Reconstruction Quality

Fréchet Inception Distance (FID), as explained in section 2.5, measures how similar reconstructed images are to real images by comparing their representations in a feature space extracted by a pretrained image recognition network. Instead of comparing images pixel by pixel, both real and reconstructed images are passed through this network, which converts them into high-level feature vectors capturing semantic content such as objects, shapes, and textures. The distribution of these feature vectors over many images is then compared between real and reconstructed data. A lower FID indicates that the reconstructed images produce feature representations that are statistically similar to those of real images, reflecting better reconstruction quality.

Codebook usage measures the proportion of discrete latent tokens that are actively used by the model during encoding. Higher usage indicates that the model effectively utilizes the available representation capacity and captures diverse patterns in the data. In contrast, low usage suggests that only a small subset of tokens is used repeatedly, which can indicate redundancy or codebook collapse.

4.3.2. Linear Probing

To evaluate the semantic and geometric quality of the learned latent representations, we utilize the linear probing setups on downstream tasks. The tokenizer is kept frozen, and the lightweight 1×1 convolutional heads are trained on top of the extracted latent feature maps. This design ensures that no additional representation learning capacity is introduced in the backbone, and performance reflects only the information encoded in the learned latent space.

1. Semantic Segmentation

For semantic segmentation, we evaluate the ability of the learned latent representations to capture high-level semantic structure using mean Intersection-over-Union (mIoU). In addition to the overall mIoU, we also report per-class IoU to analyze class-wise performance variations and understand how different semantic categories are represented in the latent space. Higher IoU values reflect better segmentation accuracy and latent token semantic quality.

2. Depth Estimation

For depth estimation, we train a linear regressor on the latent tokens to predict the depth values. This task evaluates the ability of the representations to capture geometric structure and spatial relationships within the scene. We report standard depth estimation metrics, including Root Mean Square Error (RMSE), Absolute Relative Error (AbsRel), log-scale RMSE (RMSE_log), and the accuracy metric δ_1.

Lower values of RMSE, AbsRel, and RMSE_log indicate better depth prediction accuracy, while higher δ_1 values reflect improved consistency between predicted and ground-truth depth.

For both segmentation and depth probing, we use a consistent evaluation protocol across single-frame and multi-frame setups. For temporal experiments, the backbone tokenizer expects temporally adjacent frames as input. Since the datasets consist of independent images rather than true temporal sequences, we pass the same input image twice via backbone tokenizer. This ensures all the tokenizer methods are evaluated under identical conditions.

4.3.3. Generation Quality

To assess the generative utility of the learned latent spaces, we train a downstream Diffusion Transformer (DiT) using a Flow-Matching objective. In this setup (Figure 14), an input image is first mapped by the tokenizer encoder into a sequence of continuous latent tokens. During the training phase, these tokens undergo a forward diffusion process where Gaussian noise is iteratively added to the latent representation. The DiT is then tasked with learning the vector field that predicts the velocity from the noisy latent state back toward the clean latent manifold. The DiT is trained to minimize the Mean Squared Error (MSE) between its predicted and interpolated target. During inference, the model generates novel scenes by sampling from pure Gaussian noise and following the learned trajectory to reconstruct a structured latent representation.

For generative performance evaluation, we compute the Generative Fréchet Inception Distance (gFID) by comparing the distribution of generated images to that of real images within a semantic feature space defined by a pretrained Inception-v3

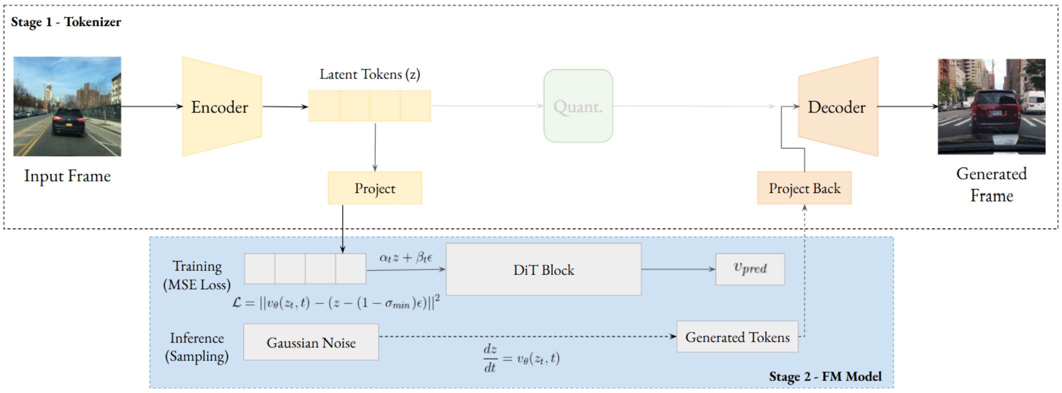


Figure 14.: Architecture: Training and inference using DiT model. During training, input frames are mapped to a 16-dimensional latent space via encoder and a projection layer. The DiT is optimized to predict the velocity of the probability flow between data and noise. At inference, the model generates novel frames by transforming Gaussian noise into latent tokens which are then projected back and decoded.

network. Specifically, the process begins with the DiT sampling from pure Gaussian noise, using the EMA-stabilized weights for optimal fidelity, and following the learned probability flow to produce clean latent tokens, which are then mapped to the pixel domain via the tokenizer decoder. Both generated and real image sets are passed through the Inception-v3 network to extract high-level feature vectors. We model these embeddings as multivariate Gaussians by estimating their means and covariance matrices, finally computing the Fréchet distance between the two distributions. A lower gFID indicates that the generated images more closely match the statistics of real images in terms of both visual quality and structural diversity.

4.4. Datasets

4.4.1. Training Data

We employ the Berkeley DeepDrive 100K (BDD100K) dataset for training. BDD100K consists of 100,000 high-resolution video clips captured across diverse cities in the

United States, under varying lighting, seasonal, and road conditions. The dataset covers a wide range of real-world driving scenarios, including urban streets, residential areas, and highways, often containing multiple interacting objects in a single frame.

For training the models we utilize the adjacent video frames from the BDD100K video dataset. To evaluate the effects of diverse data, we train our model on BDD100K image dataset, where adjacent images are not temporally correlated. We use the following two dataset configurations:

1. **Daytime Video Frames:** A subset containing only daytime video frames is used, resulting in a training set of approximately 800 videos. Each video spans around 400 frames. Videos are sampled at 10 Hz for consistency. The validation split follows the same setup, consisting of approximately 200 videos with a similar number of frames per video.
2. **Image Dataset Variant:** This configuration includes a wider variety of driving scenes, covering daytime and nighttime across multiple seasons. The training set consists of 70,000 images, and the validation set contains 1,764 images.

4.4.2. Evaluation Data

Evaluation is conducted across reconstruction quality and two linear probing setups, as described in the Evaluation Protocols section (section 4.3). The following datasets are employed for different evaluation tasks:

1. **Reconstruction Quality:** We use the BDD100K subset for reconstruction evaluation. During evaluation, rFID and codebook usage are computed using 1,000 images from the dataset.
2. **Semantic Segmentation:** We use Cityscapes dataset for semantic segmentation probing. Cityscapes contains high-resolution images captured from 50 cities in Germany, representing a diverse set of urban street scenes and is

widely used for evaluating semantic segmentation, instance segmentation, and scene understanding in real-world driving scenarios. Our configuration includes training set of 11,900 images with fine per-class annotations provided in JSON format. The validation set consists of 2,000 images from three cities, each accompanied by corresponding annotation files.

3. **Depth Estimation:** We use KITTI Driving dataset for depth estimation probing. KITTI was captured from a moving vehicle in urban, rural, and highway environments around Karlsruhe, Germany, and contains calibrated stereo video sequences along with depth maps and sensor data. KITTI serves as a standard benchmark for monocular and stereo depth estimation in autonomous driving research. For evaluation, we follow the Eigen split configuration of 4,800 frames in the training set, while 873 frames in the validation set.

4.5. Training Setup

The tokenizer and its variants are trained using PyTorch Lightning in a multi-GPU setup with NVIDIA RTX GPUs. An effective batch size of 128 is maintained with gradient accumulation. Optimization is performed using the Adam optimizer with a base learning rate of 8e-7, along with a linear warm-up over the first 5,000 steps followed by a constant schedule.

The tokenizer is initialized from a pretrained MAE model weights, trained using a ViT-B/768 backbone. The quantization bottleneck is set to 16, with a codebook size of 4096. The decoder is configured with 128 channels, two ResNet blocks, and an attention resolution of 16. For the EMA-stabilized teacher supervision setup, a teacher encoder is a copy of student encoder, maintained using an exponential moving average (EMA) with a decay of 0.996. An auxiliary weighted cosine similarity loss is applied between the student and teacher feature representations to guide training. The loss weights vary depending on the experimental setup.

All tokenizer variants are trained for approximately 20 epochs. The downstream linear probes (segmentation probe and depth probe) are trained for 60 epochs, while the image generation setup is trained for 10 epochs, with the additional decoder finetuning steps.

Implementation and Experimental Framework

Our implementation is based on the official codebase of Orbis [46]. To maintain a controlled experimental environment, we introduced targeted modifications only to the tokenizer pipeline to integrate our enriched latent objectives. Conversely, the generation pipeline remains intact. This ensures that observed results in gFID and structural coherence are directly attributable to our latent space enhancements rather than variations in the generation logic.

5. Results and Discussion

In this section, we evaluate the quality of the tokenizer representations. We assess performance using reconstruction metrics and downstream linear probe tasks. These evaluation methods measure how well the latent tokens preserve both low-level visual details and high-level semantic information. Based on these tokenizer (Stage 1) results, we identify the most effective configurations and further evaluate their performance for image generation (Stage 2).

5.1. Baseline Representation Quality

To establish a reference point, we first evaluate a baseline VQ-VAE model, trained to reconstruct the input frame. This baseline allows us to assess the contribution of additional architectural and supervisory components in improving latent representation quality.

Table 1 summarizes the performance of the VQ-VAE model. We observe that extending training beyond 10 epochs slightly improves reconstruction convergence, but leads to degradation in downstream metrics such as segmentation mIoU and depth RMSE, as shown in Figure 15. This trend indicates that optimizing for reconstruction alone does not reliably improve semantic or geometric understanding, motivating the need for more structured learning strategies in tokenizer design.

Model	rFID↓	Codebook usage ↑	Segmentation mIoU ↑	Depth RMSE ↓
VQ-VAE	29.09	15.45%	34.32	6.4156

Table 1.: Results: VQ-VAE baseline performance. Baseline performance is evaluated across reconstruction quality, codebook utilization, and linear probing tasks. rFID measures reconstruction fidelity, codebook usage reflects the number of unique codebook entries utilized, while segmentation mIoU and depth RMSE assess semantic and geometric quality, respectively.

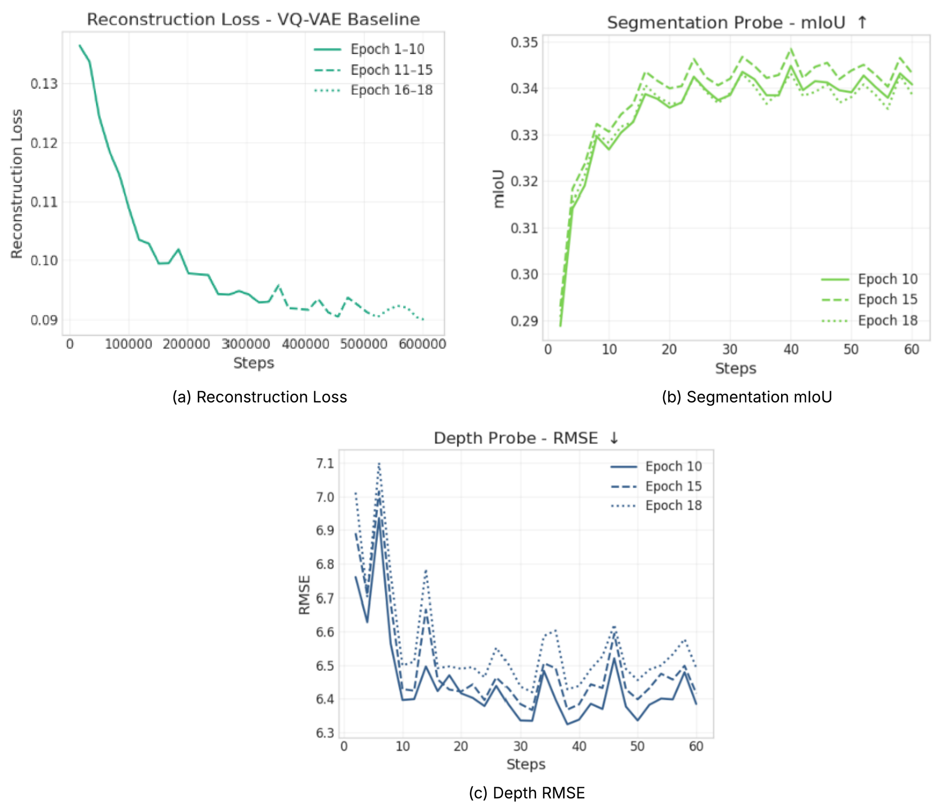


Figure 15.: Analysis: Baseline performance over extended training. (a) Validation reconstruction loss shows convergence over 10, 15, and 18 epochs of training, (b) Semantic segmentation (mIoU) shows marginal improvements for epoch 15, while degrades for epoch 18, (c) Depth estimation (RMSE) shows no performance improvement after epoch 10.

5.2. Effects of Input Configurations

The results in this section correspond to the input configuration setup shown in Figure 7, where we vary the inputs to the VQ-VAE baseline. Specifically, input configuration 1 applies spatial masking (masked image modeling), while input configuration 2 incorporates temporal context through masking (temporal context-based image modeling).

5.2.1. Masked Image Modeling

We evaluate our approach in the Masked Image Modeling (MIM) setup (subsection 4.2.1 - Input Configuration 1), where we apply different masking strategies and mask ratios to the input frame. Here, we focus on the quality of reconstruction along with the semantic and geometric performance of the model during evaluation.

Model	Mask ratio (%)	rFID↓	Codebook usage ↑	Segmentation mIoU ↑	Depth RMSE ↓
Baseline	00 - 00	29.09	15.45%	34.32	6.4156
Fixed masking	30 - 30	32.38	15.19%	34.91	6.2011
Fixed masking	50 - 50	34.62	14.65%	35.38	5.9993
Fixed masking	75 - 75	38.78	14.94%	37.31	5.9108
Range masking	00 - 50	31.60	13.94%	35.83	6.1916
Range masking	00 - 75	33.24	13.99%	35.61	6.0993
Range masking	00 - 90	32.43	13.87%	36.47	6.0660
Range masking	00 - 95	32.11	13.59%	36.01	6.1200

Table 2.: Results: Masked image modeling. Masking strategy comparison under different mask ratios. Heavy random range masking (90%) shows better balance between reconstruction fidelity (rFID), segmentation (mIoU) and depth (RMSE) in comparison to fixed masking and VQ-VAE baseline.

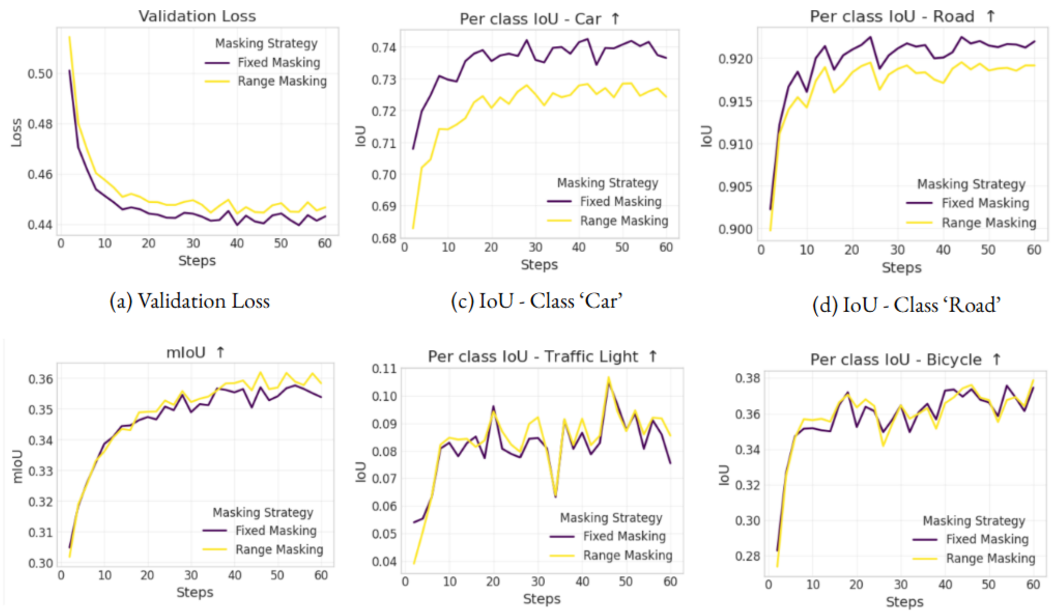
Table 2 shows that masking improves the semantic and geometric performance of the model, at the cost of marginally reduced perceptual fidelity and codebook usage compared to the VQ-VAE baseline (section 5.1). **In particular, 90% random range masking achieves the best overall balance.**

We analyze how various hyperparameter configurations, including data augmentation strategies, data diversity, and architectural modifications, impact this MIM framework. While we investigate these variations to ensure robustness, the results presented in Table 2 serve as our primary reference point and optimal configuration. Refer Appendix Table 12 for quantitative ablation.

Discussion

1. **Masking strategy influences the trade-off between reconstruction fidelity and semantic representation quality:** Random-range masking maintains stable perceptual fidelity and codebook utilization while improving segmentation and depth estimation performance at higher masking ratios. In contrast, fixed high-ratio masking yields stronger semantic and geometric gains at the cost of perceptual fidelity. This indicates that different masking strategies balance perceptual fidelity, semantic quality and geometric performance in different ways.
2. **Higher mIoU might not indicate better overall segmentation quality:** We analyze the segmentation performance of random-range and fixed masking based on the results in Table 2. Fixed masking (Figure 16) exhibits lower validation loss compared to random-range masking. Interestingly, the mIoU shows the opposite trend: despite its higher validation loss, random-range masking achieves better mIoU. This discrepancy arises from the class-averaged nature of mIoU, where each class contributes equally regardless of its pixel frequency. Per-class IoU analysis shows that random-range masking improves performance on less frequent classes (e.g., bicycles, traffic lights), whereas fixed masking per-

forms better on dominant classes (e.g., road, car). As a result, gains on rare classes disproportionately increase mIoU, even when performance on dominant classes degrades. In contrast, fixed masking better preserves global scene structure, leading to more stable segmentation performance. Overall, this indicates that mIoU alone may overestimate segmentation quality when improvements are driven primarily by less frequent classes.



(a) Validation Loss (c) IoU - Class 'Car' (d) IoU - Class 'Road'

(b) Mean IoU (e) IoU - Class 'Traffic Light' (f) IoU - Class 'Bicycle'

Figure 16.: Analysis: Per-class IoU analysis under masking strategies. Comparison of fixed and random range masking under 50% masking ratio. (a) Fixed masking shows lower validation loss, (b) despite this, fixed masking achieves poor mIoU, (c-d) dominant classes show better IoU under fixed masking, (e-f) rare classes achieve better IoU under random range masking. This indicates that range masking improves over rare classes at the cost of stability in dominant semantic structures.

5.2.2. Temporal Context-based Image Modeling

To study the impact of temporal context on representation learning, we use two adjacent video frames (x_t and x_{t+1}) as shown in Figure 7, and evaluate different masking strategies. As described in subsection 4.2.1 - Input Configuration 2, we consider four masking setups:

1. **Full context:** Both input frames (x_t and x_{t+1}) are fully visible.
2. **Single-frame mask:** Frame x_t is fully visible and serves as context, while frame x_{t+1} is masked with a 50% masking ratio.
3. **Asymmetric mask:** Both frames are masked, with frame x_{t+1} subjected to 50% masking and frame x_t subjected to 30% masking. Tube masking is applied, meaning the same spatial patches are masked across both frames.
4. **Heavy range mask:** Both frames are heavily masked using a 90% random-range masking strategy.

In all configurations, frame x_t serves as context, while only the tokens from frame x_{t+1} are vector-quantized and decoded. Here we compare the performance of the model with two baselines: **VQ-VAE baseline** (baseline 1 - Table 1), to assess the impact of temporal context on representation quality, and **MIM baseline with 90% masking** (baseline 2 - Table 2), to evaluate how temporal context-based masking compares to spatial masking.

Setting	Masking ratio (%)		rFID↓	Codebook usage ↑	Segmentation mIoU ↑	Depth RMSE ↓
	F1	F2				
Baseline 1	00	NA	29.09	15.45%	34.32	6.4156
Baseline 2	90	NA	32.43	13.87%	36.47	6.0660
Full context	00	00	32.41	15.20%	32.08	7.5976
Single-frame mask	00	50	30.62	14.57%	32.71	6.2427
Asymmetric mask	30	50	33.11	14.20%	32.69	6.9898
Heavy range mask	90	90	31.86	13.72%	35.46	6.1020

Table 3.: Results: Temporal context-based image modeling. Comparison across different masking strategies and baselines (Baseline 1 - VQVAE, Baseline 2 - Best MIM setup from Input Config. 1 (Table 2)). 'Heavy Range Mask' shows improved semantics gains over Baseline 1, while remains semantically and geometrically inferior to Baseline 2.

In Table 3, we observe that incorporating temporal context without appropriate masking leads to degraded performance across all evaluation metrics. While 'Single-frame mask' improves reconstruction quality over the 'Full context' setting, it shows poor semantic and geometric quality. 'Asymmetric mask' further deteriorates the performance. In contrast, **'Heavy range masking' applied to both frames achieves a better balance across all evaluation metrics, providing additional semantic and geometric gains over the standard VQ-VAE baseline (baseline 1). However, its semantic and geometric performance remain inferior to spatial masking (baseline 2).**

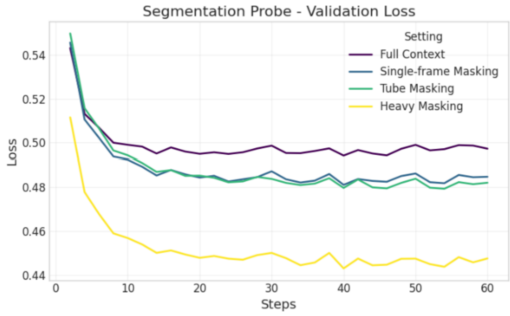
Discussion

1. **Heavy random range masking can effectively leverage temporal context:** The results in Table 3 show that 'Full context' and 'Single-frame mask' settings enables information copying, leading to faster convergence and lower reconstruction loss (Figure 17). However, this introduces shortcut learning and limits the effectiveness of temporal information on the semantic and geometric quality. 'Asymmetric masking' (Tube Masking) creates imbalance across frames, making reconstruction and depth estimation more difficult due to lack of reliable anchors. 'Heavy range masking' applies balanced corruption to both frames, preventing shortcuts and promoting consistent spatiotemporal learning, resulting in improved overall performance across metrics

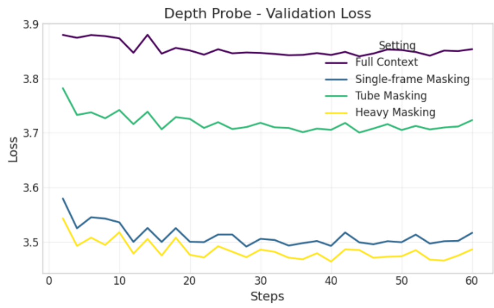
2. **Masking induces more compact codebook usage with improved semantic representations:** We observe that introducing masking consistently reduces codebook usage across all settings compared to VQ-VAE baseline, while simultaneously improving downstream semantic and geometric performance. This indicates that the model shifts towards a more compact subset of latent codes as reconstruction becomes more challenging, prioritizing global and semantically meaningful structure over fine-grained token diversity. In our

setting, we monitor codebook usage to ensure stability, which remains consistently in the range of 13–15% across all masking configurations, with no signs of collapse observed.

(a) Reconstruction Loss



(b) Segmentation Loss



(c) Depth Loss

Figure 17.: Analysis: Validation loss convergence under temporal context-based image modeling. 'Full context' and 'Single-frame masking' show lower reconstruction and high probe losses (segmentation and depth loss) due to information copying, 'Asymmetric masking' (Tube masking) shows overall instability, 'Heavy masking' on both the frames shows better loss convergence and hence better balance across all evaluation metrics.

Key Takeaway:

For VQ-VAE with modified input configurations, heavy random range masking helps balance local (perceptual) and global (semantic and geometric) representation quality. While both spatial and temporal masking learn useful representations (Table 3), spatial masking (Table 2) offers a consistent balance between semantic and geometric quality along with perceptual fidelity.

5.3. Impact of Auxiliary Supervision on Student Features

We evaluate the effect of incorporating knowledge-distillation as an auxiliary supervision signal for training the tokenizer, as explained in approach Figure 9. In this setup, the student features are aligned with features from a frozen pretrained teacher network, and we analyze its impact on perceptual, semantic and geometric quality of the latent tokens.

Here we consider four pretrained teacher models: DINO, Depth-Anything v2, RAFT and self-distillation from our pretrained MIM model (90% spatial masking setup - Table 2). We compare the performance of these models against the VQ-VAE baseline.

Model	rFID↓	Codebook usage ↑		Segmentation mIoU ↑	Depth RMSE ↓
		CB rec	CB aux		
Baseline	29.09	15.45%	NA	34.32	6.4156
DINO distill.	29.60	14.82%	97.85%	44.34	5.9185
Depth distill.	30.70	15.52%	42.50%	38.39	6.1160
RAFT distill.	30.87	15.53%	42.53%	36.10	6.0661
Self distill.	35.16	12.69%	100%	27.54	6.8400

Table 4.: Results: Auxiliary supervision via knowledge distillation. Here 'CB rec' refers to codebook from reconstruction branch and 'CB aux' refers to the codebook from distillation branch. All pretrained teacher models (DINO, Depth, RAFT) consistently improve reconstruction and downstream probe performance over VQ-VAE baseline. Self-distillation using our pretrained MIM model (Self distill.) as the teacher performs weaker under the same architecture setting.

From Table 4 we observe that auxiliary supervision from pretrained teacher models, like DINO, Depth and RAFT, improves the semantic and geometric performance (segmentation and depth) of the model, at the cost of marginal degradation in the perceptual quality. This shows that **auxiliary supervision improves semantic**

performance when compared to the baseline and its impact depends on the choice of teacher model. Using self-distillation with our best setting (random range masking 90%) from pretrained MIM model (Table 2), shows overall poor performance under same architectural setting as other pretrained teacher models in Table 4. This indicates that the influence of teacher model plays a key role in this architectural setting.

To evaluate the impact of our pretrained MIM model, we modified the distillation branch to provide supervision in the continuous latent space. While this configuration yielded significant gains over the self-distillation results in Table 4, it did not surpass VQ-VAE baseline. These results suggest that while the benefits from a pretrained MIM model are more evident under this modified architecture, the overall performance remains limited. Architectural details and results are provided in the Appendix Figure 27 and Table 14.

Additionally, we investigate the impact of temporal context for this auxiliary supervision via knowledge distillation setting by considering temporal context (frame x_t and x_{t+1}) as input. We observe that incorporating temporal context in this setting shows comparatively poor performance (see Appendix Table 13 and Figure 26 for results and architecture).

Discussion

1. **Task-specific supervision does not necessarily translate to superior task-level gains:** The differences across teacher models can be attributed to the nature of their supervision signals. DINO provides semantically rich and globally consistent features, which lead to stronger performance across all the evaluation metrics. In contrast, Depth and RAFT offer more specialized cues, geometric structure and motion, respectively, resulting in more moderate improvements. Notably, although the depth model is explicitly trained to capture

depth cues, it does not outperform DINO-based supervision, even on depth estimation task. This suggests that general semantic representations learned by DINO provide a stronger supervisory signal than task-specific features from Depth and RAFT models.

2. **Effect of design choice on self-distillation:** In the self-distillation setup, we observe that replacing the quantizer from the distillation branch with a lightweight predictor, along with an appropriate masking strategy, demonstrates comparatively better performance. This suggests that supervision from a pretrained model in a continuous latent space avoids the limitations of discretization. Masking further provides a meaningful learning signal. Together, these design choices lead to more stable and informative representation learning when using our pretrained MIM model as the teacher.

Key Takeaway:
Knowledge distillation demonstrates a clear positive effect in achieving more balanced and improved representation quality across all the evaluation metrics, especially when the teacher is a strong pretrained model like DINO, Depth or RAFT. Incorporating temporal context in this setting shows poor performance across all the evaluation metrics.

5.4. Effects of Cross-frame Temporal Reconstruction

We evaluate a lightweight cross-frame temporal reconstruction extension for Input Configuration 2 setup (subsection 4.2.1). Here latent tokens from a masked target frame (x_{t+1}) are used to reconstruct both context frame (x_t) and target frame (x_{t+1}), see Figure 10.
Additionally, we verify the robustness of this design by applying DINO-based auxiliary supervision as an additional experiment. Here we compare the performance of the model against a simple VQ-VAE baseline.

Model	rFID↓		Codebook usage ↑	Segmentation mIoU ↑	Depth RMSE ↓
	F1	F2			
Baseline	29.09	NA	15.45%	34.32	6.4156
Temporal compress.	31.54	32.72	15.19%	32.96	6.2104
+ DINO supervision	30.48	31.77	14.84%	43.06	5.6800

Table 5.: Results: Cross-frame temporal reconstruction. The model uses two input frames, with the target frame masked at 50% ratio. Temporal compression with DINO distillation (last row) shows improved performance in comparison to the VQ-VAE baseline. Note: DINO supervision (last row) is a distillation setup and hence maintain aux branch codebook with usage of 96.19%.

From Table 5, we observe that **temporal compression shows geometric improvements (RMSE) over VQ-VAE baseline. Introducing DINO supervision improves the overall performance of the model.** We achieve these semantic and geometric gains at the cost of marginal degradation in perceptual quality in comparison to the baseline.

The architectural design for cross-frame temporal reconstruction with DINO supervision can be found in Appendix Figure 28.

Discussion

Context frame dominates reconstruction in cross-frame temporal reconstruction: We observe that cross-frame reconstruction yields higher perceptual fidelity for the context frame compared to the target frame. This happens because the context frame is fully visible and acts as a strong anchor, while the partially masked target frame is harder to reconstruct, leading the model to rely more on context information. Continued training improves reconstruction for both frames, but it comes at the cost of degraded semantic performance, indicating a shift toward pixel-level optimization. Figure 18 shows the reconstruction loss convergence for

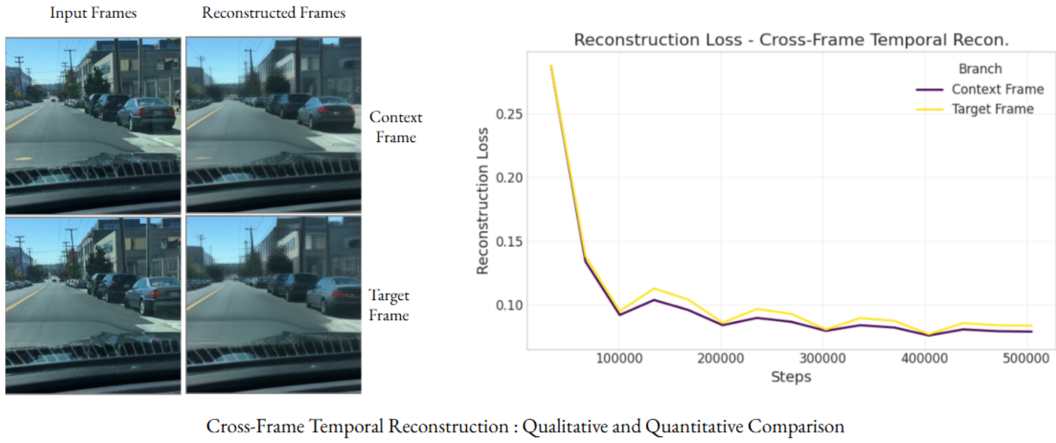


Figure 18.: Analysis: Visualization of reconstructed frames and corresponding loss under cross-frame temporal reconstruction. Qualitative and Quantitative comparison of context and target frame reconstruction in the cross-frame temporal reconstruction setting. The context frame achieves higher perceptual fidelity despite tokens being derived from the target frame, highlighting the effect of input visibility and masking on reconstruction behavior.

context and target frame, with context frame showing lower reconstruction loss and better perceptual fidelity.

Kay Takeaway:

Although cross-frame temporal reconstruction introduces a more challenging learning objective, it yields improved geometric performance at the cost of marginal degradation in the reconstruction fidelity. When paired with DINO distillation, we experience a positive improvement in overall performance in this setting.

5.5. Choice of Supervision Space for MAE

We evaluate the impact of supervision from an EMA-stabilized teacher encoder in the MAE setup, along with the choice of supervision location - direct supervision (continuous latent space - Figure 11) or indirect supervision (discrete latent space - Figure 12, decoder feature space - Figure 13). The goal is to isolate how different supervision designs influence representation quality and reconstruction performance. We compare the performance of these settings against a VQ-VAE baseline. We finally evaluate the best-performing tokenizer configuration on image generation task.

5.5.1. Continuous Latent Space Supervision

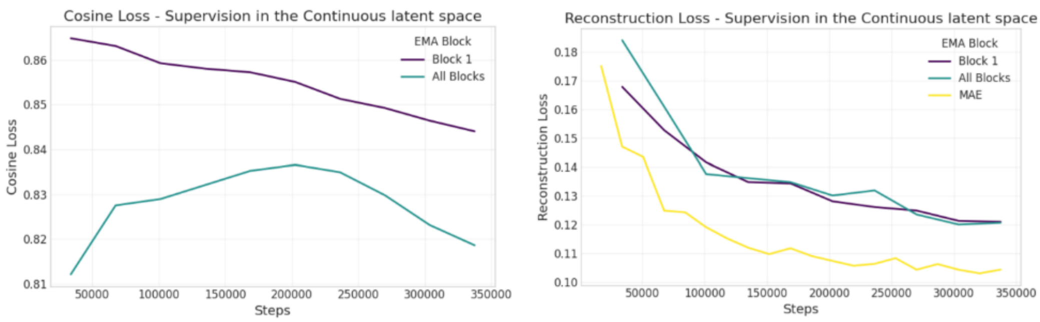
We begin with the continuous latent space setting, where supervision is applied before quantization (Figure 11). In all experiments, we use a fixed masking ratio of 0.5, a cosine loss weight (λ_{cosine}) of 0.2, and a full reconstruction loss weight (λ_{recon}) of 1.0. This controlled configuration allows us to study the impact of continuous feature alignment while keeping the VQ-VAE reconstruction objective unchanged.

Model	EMA block	rFID↓	Codebook usage ↑	Segmentation mIoU ↑	Depth RMSE ↓
Baseline	NA	29.09	15.45%	34.32	6.4156
	Block 1	34.31	13.15%	33.09	7.4240
	All blocks	35.74	12.93%	33.37	7.5940

Table 6.: Results: EMA-stabilized teacher supervision in continuous latent space. ‘EMA Block’ denotes the teacher encoder block used for cosine similarity alignment with student features. All student encoder blocks are used for feature extraction. Both blocks show competitive results when compared to each other. However, this approach remains inferior to the Baseline.

As shown in Table 6, the features from ‘Block 1’ of EMA-stabilized teacher model shows better rFID, codebook usage and geometric performance in comparison to ‘All blocks’ setting, while ‘All blocks’ setting shows better segmentation mIoU. These results suggest that supervision from intermediate and last block of the EMA-supervised teacher model shows competitive results, however they indicate a poor performance, when compared to the VQ-VAE baseline.

1. **Deeper features are harder to align, leading to unstable optimization, even if final performance looks similar:** As shown in Figure 19 supervision from 'All blocks' exhibits less stable cosine loss dynamics compared to 'Block 1', despite similar final reconstruction performance. This suggests that deeper features from all blocks provide a noisier supervision signal, whereas shallow features from block 1 enable more reliable alignment. Additionally, even with a low cosine loss weight ($\lambda_{cosine} = 0.2$), we observe slight degradation in reconstruction quality, indicating a potential objective mismatch between reconstruction and feature alignment.



(a) Cosine Loss

(b) Reconstruction Loss

Figure 19.: Analysis: Cosine loss dynamics for 'Block 1' vs 'All Blocks' supervision: (a) 'Block 1' exhibits stable convergence, 'All Blocks' shows initial divergence, while converges slowly at the later stage, (b) Both the 'Block 1' and 'All Blocks' setting converge at the same point in reconstruction loss. Even with the $\lambda_{cosine} = 0.2$ and $\lambda_{recon} = 1.0$ for this supervision method, we notice a significant difference for the reconstruction convergence in comparison to MAE ($\lambda_{cosine} = 0.0$,$\lambda_{recon} = 1.0$).

2. **Impact of EMA-stabilized supervision in continuous latent space on the geometric quality of the tokens:** The consistently poor depth performance indicates that the proposed supervision strategy does not preserve the fine spatial structure necessary for geometric reasoning.

5.5.2. Discrete Latent Space Supervision

Discrete latent space supervision is implemented with a cosine similarity loss applied on the quantized representations (Figure 12), while maintaining the standard reconstruction objective. We implement this setup under masked inputs, following the same masking protocol as the continuous latent space setting - fixed masking with mask ratio 0.5, and $\lambda_{cosine} = 1.0$.

Model	λ_{recon}	Masking	rFID ↓	Codebook usage ↑	Segmentation mIoU ↑	Depth RMSE ↓
Baseline	1.0	✗	29.09	15.45%	34.32	6.4156
	0.20	✓	54.47	4.00%	21.30	7.1517
	0.20	✗	41.54	13.94%	26.16	6.8018
	1.0	✗	36.74	14.31%	29.92	6.5596

Table 7.: Results: EMA-stabilized teacher supervision in discrete latent space. The second row corresponds to masked input with ratio of 0.5, where we observe codebook collapse. The settings without masking shows monotonic improvement across all the evaluation metrics on increased reconstruction loss weight. However, this setting remain inferior to the VQ-VAE Baseline.

Table 7 shows that masking leads to unstable optimization and a sharp drop in codebook utilization, indicating difficulty in learning meaningful discrete representations. Removing masking stabilizes training and significantly improves codebook usage. Furthermore, in the unmasked setting, increasing reconstruction loss weight consistently improves performance with the trend of monotonic improvement observed (Appendix Table 15), highlighting the stabilizing effect of reconstruction loss. **Overall, these results suggest that combining masking with supervision in a quantized latent space introduces challenging optimization dynamics.**

To further understand if these observations are design specific, we decouple the quantizer in a separate branch and analyze the performance (architecture design can be found in Appendix Figure 29).

Setting	rFID↓	Codebook usage ↑		Segmentation mIoU ↑	Depth RMSE ↓
		CB rec	CB aux		
Unmasked	36.02	14.50%	100%	28.06	6.6912
Masked	59.69	10.72%	100%	18.60	9.8913

Table 8.: Results: EMA-stabilized teacher supervision in discrete latent space (decoupled quantizer setting). Model is trained with unmasked and masked input configurations. Masked input follows a ratio of 0.5 for easy comparison with the results from Table 7. We maintain $\lambda_{cosine} = 1.0$ and $\lambda_{recon} = 1.0$ since loss weighing doesnot show any differences in the previous setting. Despite additional flexibility in this setting, the improvements are hardly observed.

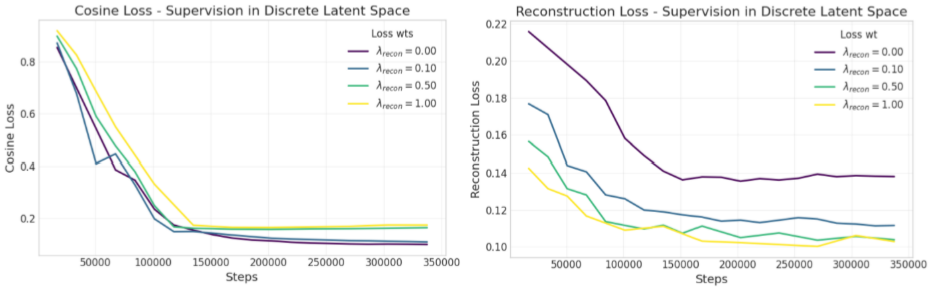
Table 8 shows that the codebook collapse observed in Table 7 for masking is resolved. However, improved codebook utilization in the auxiliary branch does not translate to better overall performance. While reconstruction improves slightly in the unmasked setting, segmentation and depth performance degrade. **This indicates that supervision in the discrete token space does not inherently improve representation quality, even under the more flexible configuration.**

Discussion

1. **Masking introduces optimization instability in discrete space:** Masking increases uncertainty in the latent features. When these uncertain features are quantized into discrete codebook entries, even small variations can lead to different token assignments, making training unstable. At the same time, the model must balance two competing objectives: reconstructing fine-grained visual details and aligning with teacher supervision that emphasizes higher-level semantics. As a result, the model tends to rely on a small subset of frequently

used ‘safe’ tokens, leading to poor codebook utilization and a decline in reconstruction quality.

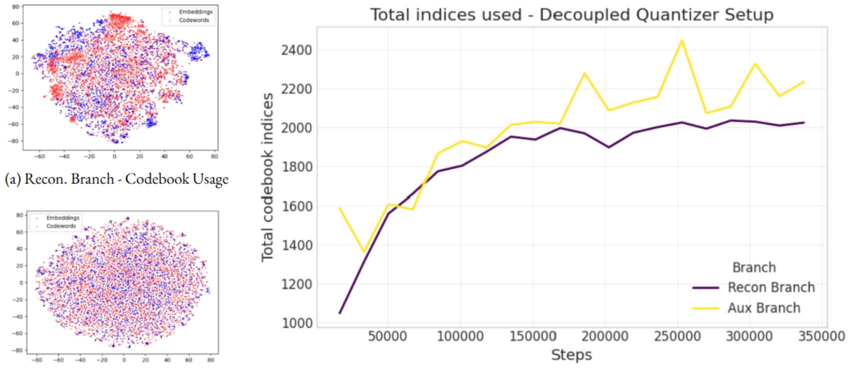
2. **Reconstruction is the stabilizing force:** Reconstruction serves as the primary stabilizing signal for discrete latent supervision. Increasing reconstruction loss weight consistently improves codebook utilization and downstream performance (Table 7). As shown in Figure 20, higher reconstruction loss weights accelerate reconstruction convergence by marginally affecting cosine loss convergence, explaining why stronger reconstruction does not lead to objective mismatch in this setting.



(a) Cosine Loss (b) Reconstruction Loss

Figure 20.: Analysis: Reconstruction and Cosine loss under EMA-stabilized teacher supervision in the discrete latent space. The figure illustrates the convergence of reconstruction and cosine losses during validation. (a) Increasing the reconstruction loss weight has only a marginal effect on cosine loss convergence. (b) A higher reconstruction loss weight leads to improved reconstruction loss convergence.

3. **Codebook usage requires careful interpretation:** Codebook usage is a key diagnostic for discrete representation learning but must be interpreted per branch. As shown in Table 8 and Figure 21, the reconstruction branch utilizes 14.5% of the codebook, while the auxiliary branch achieves 100% coverage. However, the total number of indices used during validation remains similarly low for both branches. Therefore, the high codebook usage reported for the auxiliary branch in Table 8 cannot be considered optimal. Although it activates



(a) Recon. Branch - Codebook Usage

(b) Aux. Branch - Codebook Usage

(c) Total indices used during validation

Figure 21.: Analysis: Codebook usage in a decoupled quantizer (Figure 29, Results Table 8) under EMA-stabilized teacher supervision in the discrete latent space. (a) Codebook usage for the quantizer in the reconstruction branch which has the reconstruction-only objective, (b) Codebook usage for the quantizer in the semantic alignment (aux) branch, (c) Total number of indices used during the validation. Although the auxiliary branch shows 100% codebook coverage, as indicated by t-SNE visualizations and quantitative results in Table 8, the actual number of indices used during validation remains low, suggesting that this coverage does not reflect meaningful or consistent utilization.

all codebook entries, this does not necessarily reflect meaningful or consistent usage.

5.5.3. Decoder Feature Space Supervision

We next explore feature-level alignment in the decoded feature space (Figure 13), where supervision operates on post-decoder representations and corresponds to higher-level semantic features. In this setting, we fix $\lambda_{cosine} = 0.2$ and $\lambda_{recon} = 1.0$, with the mask ratio of 0.5, similar to the continuous (subsection 5.5.1) and discrete space (subsection 5.5.2) supervision settings.

We observe in Table 9, that supervision in the decoded feature space achieves competitive segmentation and geometric performance while maintaining stable codebook

Model	λ_{cosine}	rFID↓	Codebook usage ↑	Segmentation mIoU ↑	Depth RMSE ↓
Baseline	NA	29.09	15.45%	34.32	6.4156
	0.2	33.52	15.04%	35.25	6.1494
	0.5	34.06	15.58%	34.49	6.0638
	1.0	34.87	16.18%	34.05	6.0777

Table 9.: Results: Feature-level supervision in decoded feature space. Feature level supervision under fixed $\lambda_{recon} = 1.0$ and mask ratio of 0.5. We vary the cosine loss weight to analyze it impact across the evaluation metrics. Model with $\lambda_{cosine} = 0.2$ shows best overall semantic and geometric performance.

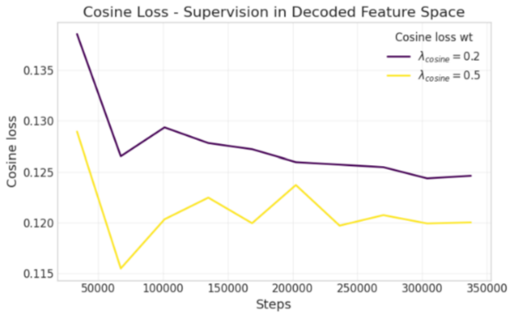
utilization. In particular, the model with $\lambda_{cosine} = 0.2$ outperforms the VQ-VAE baseline in terms of semantic and geometric gains. However, further increasing the cosine loss weight leads to degraded performance.

Overall, these results suggest that supervision in the decoded feature space improves both segmentation and depth performance at a cosine loss weight of 0.2. Moreover, alignment in the decoder feature space provides a more balanced trade-off compared to supervision in continuous or discrete latent spaces.

Discussion

Decoder feature-space provides stability with weak cosine alignment: The performance observed at a lower cosine loss weight ($\lambda_{cosine} = 0.2$) suggests that only mild feature alignment is beneficial in the decoded space, where representations are already semantically structured. A small cosine loss weight acts as a regularizer, preserving semantic consistency while allowing reconstruction to dominate. In contrast, higher cosine loss weights over-constrain the features, degrading reconstruction performance along with segmentation and depth performance (Figure 22).

Key Takeaway:



(a) Cosine Loss

(b) Reconstruction Loss

Figure 22.: Analysis: Reconstruction and cosine loss for feature alignment in the decoder feature space. (a) A higher cosine loss weight leads to faster convergence of the cosine loss, while (b) it makes reconstruction loss convergence slightly more difficult. This suggests that decoder feature-space alignment is more stable at lower cosine loss weights.

From the EMA-stabilized teacher supervision setups described above, we find that feature alignment is more effective in the decoded feature space than in continuous or discrete latent spaces. Our results indicate that applying supervision in the decoded feature space achieves a better balance between semantic and geometric richness along with local details.

5.6. Evaluation - Image Generation

We evaluate the effectiveness of the learned representations from tokenizer (Stage 1) on image generation task using a Diffusion Transformer (DiT). We analyze how various tokenizer designs impact final generative performance. Our study focuses on the trade-off between latent token quality and downstream generative performance.

During this evaluation we select the best configurations from each of our tokenizer experiments that demonstrate improved semantic performance over VQ-VAE baseline. We list the selected configurations as follows:

1. **VQ-VAE:** Standard VQ-VAE baseline model (Figure 6)

2. **MAE:** Input Configuration 1 with random range masking (Mask ratio: 90% - Figure 7)
3. **Temporal MAE:** Input Configuration 2 with 'Heavy Range Mask' setting (Mask ratio: 90% - Figure 7) and temporal context
4. **DINO distill.:** Auxiliary Supervision with DINO pretrained prior (Figure 9)
5. **Depth distill.:** Auxiliary Supervision with Depth pretrained prior (Figure 9)
6. **RAFT distill.:** Auxiliary Supervision with RAFT pretrained prior (Figure 9)
7. **EMA supervision:** Supervision in the decoded feature space with fixed masking (Mask ratio: 50% - Figure 13)

By using these configuration, we analyze how the representations from a multilevel tokenizer affect the image generation quality.

Model	Tokenizer	rFID↓	gFID↓	Segmentation mIoU ↑	Depth RMSE ↓
Baseline	VQ-VAE	29.09	45.09	34.32	6.4156
Ours	MAE	32.43	50.88	36.47	6.0660
Ours	Temporal MAE	31.86	55.10	35.46	6.1020
Ours	DINO distill.	29.60	44.03	44.34	5.9185
Ours	Depth distill.	30.70	44.60	38.39	6.1160
Ours	RAFT distill.	30.87	45.31	36.10	6.0661
Ours	EMA supervision	33.52	49.29	35.25	6.1494

Table 10.: Results: Image generation evaluation. We consider best tokenizer settings for evaluation. From the above results, all the tokenizers show improved semantic and geometric quality of latent tokens, while auxiliary supervision from pretrained priors (DINO, Depth) show improved generation performance over the baseline VQ-VAE model.

Table 10 and scatter plot (Figure 23) illustrate how the representation quality of latent tokens correlates with the final generative performance of the Diffusion Transformer (DiT). Methods incorporating auxiliary supervision achieve the best generative performance, along with strong representation quality. In contrast, we observe that approaches such as MAE, temporal MAE, and EMA-based supervision exhibit a balanced trade-off between perceptual quality along with semantic and geomet-

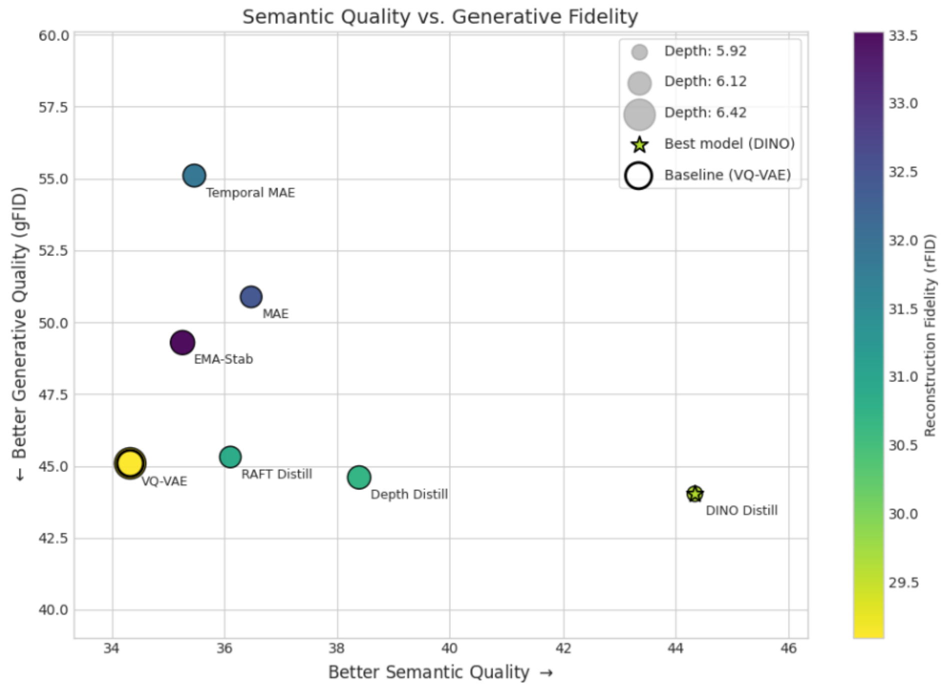


Figure 23.: Analysis: Evaluating the trade-off between representation quality and generative fidelity. This analysis evaluates the trade-off between representation quality (rFID, mIoU, RMSE) and generative fidelity (gFID) across various tokenizer configurations. DINO-based distillation achieves the best overall performance, while methods such as MAE, although semantically rich, produce lower-fidelity generated images.

ric performance (high mIoU and low RMSE). However, despite these improvements in representation quality, they do not achieve competitive generative performance compared to the baseline VQ-VAE model.

Discussion

Auxiliary supervision from pretrained priors: Tokenizers trained with strong pretrained priors such as DINO, Depth, and RAFT demonstrate overall balanced performance. These pretrained models provide substantial improvements in the semantic and geometric quality of the latent tokens, with marginal degradation in perceptual fidelity. While distillation from semantic and geometric priors such as DINO and Depth, improve both representation quality and image generation performance, RAFT-based distillation yields comparatively weaker gains. This may be attributed to the motion-centric nature of RAFT supervision, which emphasizes temporal correspondence rather than static semantic structure, resulting in less effective latent organization for image generation tasks.

MAE and Temporal MAE: MAE and Temporal MAE with random-range masking (90% mask ratio) achieve strong representation quality (rFID, mIoU, RMSE) but result in poor generative quality. Despite stable convergence in DiT validation loss (Figure 25), the model produces limited high-quality novel scenes, indicating that improvements in the training objective do not necessarily translate to better generative performance. Further analysis shows that fixed masking (Table 11) improves semantic quality and gFID, whereas random masking introduces variability and degrades performance.

Hypothesis - Does high mIoU in masking setup necessarily imply that a latent space contains the semantic information relevant for image generation?

Our analysis suggests that range masking, while achieving competitive mIoU, does not adequately encode the critical structural features of the driving environment, as discussed in results section 5.2. Per-class segmentation analysis (Figure 24) confirms that range masking exhibits weaker performance on dominant classes in our dataset, such as truck, bus, car, and building. In contrast, it improves performance on smaller or less frequent classes, including person, fence, traffic light, and bicycle. As mIoU is averaged across classes, improvements on these underrepresented classes can lead to an overall increase in the metric, even when performance on dominant structural classes degrades. This explains why range masking achieves higher mIoU despite showing poorer performance on the most structurally significant scene elements.

This might be the reason why random-range masking lacks the structural consistency required for high-fidelity generation. Conversely, fixed masking, even at aggressive

Model	Mask Ratio (%) low - high	rFID↓	gFID↓	Segmentation mIoU ↑	Depth RMSE ↓
Baseline	00 - 00	29.09	45.09	34.32	6.4156
Range masking	00 - 90	32.43	50.88	36.47	6.0660
Range masking	00 - 75	33.24	55.76	35.61	6.0993
Range masking	00 - 50	31.60	49.08	35.83	6.1916
Fixed masking	75 - 75	38.78	46.75	37.31	5.9108
Fixed masking	50 - 50	34.62	49.09	35.38	5.9993
Fixed masking	30 - 30	32.28	50.58	34.91	6.2011

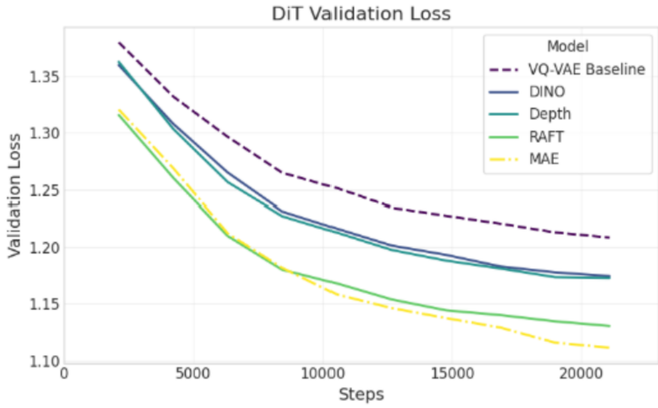
Table 11.: Results: Image generation under varied mask ratios and masking strategies. Masking ratio and masking strategy in MAE significantly influence the generative quality of the learned latents. High-ratio fixed masking (75–75) improves the semantic quality of the tokens, which in turn leads to better gFID.

mask ratios like 75%, maintains high IoU for dominant classes. This preservation of structural information translates to better generative performance compared to the range-masking variants. This suggests that for generative tasks, **stability in masking is more critical than the variability introduced by random range masking.**

Further, the **EMA supervision model** (Table 10), despite weaker semantic metrics, achieves better gFID than MAE setting. This is likely due to its fixed 50% masking ratio, which preserves token distribution stability. This suggests that stable, moderate masking is more conducive to generative performance than aggressive or variable masking strategies.

Loss convergence of DiT model: The training loss convergence patterns (Figure 25) reveal consistent difference across tokenizers. The VQ-VAE exhibits relatively higher loss, while models such as MAE, Depth and RAFT achieve lower loss values during the training. These differences suggest that certain latent spaces are easier for DiT to model under the denoising objective. However, our results show that lower loss does not translate to better generative performance.

Figure 24.: Analysis: Per class IoU for fixed and range masking comparison. Left: Dominant structural classes (Truck, Bus, Car, Building) show superior performance under Fixed Masking 0.75, **Right:** Rare classes (Person, Fence, Traffic Light, Bicycle) often exhibit higher IoU under Range Masking 0.9 and 0.75. This divergence helps explain why higher mIoU in range-masked models does not necessarily translate to better generative fidelity, as the model shows lower mIoU for most frequent classes in the dataset.



DiT Validation Loss - Convergence Comparison

Figure 25.: Analysis: DiT loss convergence. Validation loss curves across different tokenizer settings. The standard VQ-VAE exhibits the highest loss. In contrast, MAE and pretrained models converge to lower values, indicating more predictable latent representations. DINO represents a middle ground with semantically richer but more complex features.

Despite this optimization efficiency, MAE yields inferior gFID scores. In contrast, pretrained teacher-based models (DINO, Depth, RAFT) and the VQ-VAE baseline exhibit better-organized latent spaces that align more effectively with the generative objective.

Is it beneficial to balance perceptual and semantic objectives?

Balancing reconstruction and semantic objectives is beneficial when the latent space remains well-structured. Our results show that jointly optimizing all three evaluation metrics (rFID, mIoU, and RMSE), can improve semantic and geometric quality. However, masking-based objectives often introduce irregularities in the latent space, leading to less consistent representations that are harder for DiTs to model during generation. As a result, these representations are less suitable for high-quality image generation. In contrast, approaches such as auxiliary supervision (distillation) produce more structured and consistent latent representations, yielding stable improvements in DiT based image generation. Overall, this suggests that generative performance depends more on the consistency and continuity of the latent space than only on the semantic richness of individual tokens.

6. Conclusion and Future Work

6.1. Conclusion

In this thesis, we investigate how different tokenizer designs affect multilevel representation learning, with a focus on various self-supervised learning (SSL) approaches to enrich the continuous latent space of the tokenizer. By balancing the semantic and perceptual quality of the latent tokens, we develop a more robust tokenizer that improves both discriminative understanding and generative modeling performance.

Our methodology involved an extensive review of the self-supervised learning (SSL) literature, enabling us to integrate both traditional and modern feature-learning techniques with our proposed architectural modifications. We experimented with various input configurations, including spatial and temporal masking, distillation-based frameworks, temporal compression settings, and hybrid loss functions, to move beyond standard reconstruction-focused training.

The experimental results across these approaches demonstrate consistent improvements in multilevel representation quality. In each experimental branch, we identify configurations that yield measurable gains in semantic benchmarks over the standard VQ-VAE baseline, confirming that controlled masking and strategic supervision can effectively inject high-level semantic information into continuous latent tokens.

Finally, one of our primary configurations achieved a superior balance, offering higher semantic richness (29.2% improvement), stable geometric quality (8.4% improvement) and better image generation fidelity (2.4% improvement) than traditional baseline. While all tokenizers demonstrate strong semantic, geometric and perceptual stability, the results suggest that generative modeling benefits most from latent spaces supervised with strong semantic priors. Masking-based objectives yield competitive tokenizer performance; however, these improvements translate only marginally to generative performance. This indicates that generation quality is influenced not only by the balance between perceptual and semantic properties of the latent representations, but also by the nature of the supervision signal and the training objective.

6.2. Future Work

Our research demonstrates that balancing the semantic and perceptual qualities of a tokenizer is a non-trivial challenge. While our approaches achieve this balance, they also reveal that downstream generative performance depends strongly on the supervision signal and training objectives used during tokenizer learning.

Building on these findings, future work could focus on:

Advanced Masking Strategy: While the random and fixed masking strategies explored in this work are simple and effective, they may limit overall generative performance. Modern frameworks such as I-JEPA [47], employ multi-block masking strategies combined with a prediction objective. Although this approach has shown strong performance on ImageNet dataset in non-generative settings, its effectiveness on complex driving datasets, particularly when combined with reconstruction-based objectives such as our MIM setup (subsection 4.2.1) remains an open question. Overall, adopting alternative masking strategies for tokenizer training may further improve downstream generative performance.

Integration of Contrastive Objectives: While our work focused on EMA-stabilized teacher supervision on masked input (subsection 4.2.4), the role of contrastive learning within this architecture remains unexplored. In particular, viewinvariance-based contrastive supervision could potentially encourage a more structured and discriminative latent space.

Multi-Layer Auxiliary Supervision: Our current framework utilizes supervision from the final layer of the teacher model. Exploring multi-block supervision could further enrich the latent representations. Similar idea has been explored by the V-JEPA 2.1 [48], Bootleg [40] and our 'EMA-supervision in continuous latent space (subsection 4.2.4)' setting. However, incorporating multiple intermediate layers with the downstream reconstruction objective is an open direction to explore.

Structured Loss Function: Another promising direction is representation learning with structured loss functions that go beyond uniform token-wise supervision and instead apply spatially or semantically informed weighting schemes such as DINOv3's [49] global anchoring loss and JEPA's [48] context loss weighing scheme. Analyzing the impact of these loss functions individually in contrastive and masked supervision setups would be an interesting direction to explore.

Overall, this work provides a systematic study of tokenizer design and its role in shaping representations for DiT-based image generation. While the approaches discussed above remain an open direction, our results establish key insights into the trade-offs between semantic structure, perceptual fidelity, and generative performance, forming a foundation for future research in this area.

A. Appendix

This appendix provides additional experimental data and ablation studies that support the findings presented in the main text. Specifically, it includes extended experiments, additional architecture modifications and detailed analysis of the design choices made during the development of the multilevel tokenizer.

A.1. Masked image modeling across varied hyperparameter settings

This section discusses the robustness of different hyperparameter settings in the Masked Image Modeling (MIM) framework (subsection 4.2.1 - Input Configuration 1). We evaluate these hyperparameters using a fixed masking ratio of 0.5 and present the results in Table 12.

Hyperparameter details:

- Data augmentation: Random resize center
- Diverse frame data: Frames from 1600 videos (each video contains 400 frames)
- Diverse image data: 70K images, spanning across diverse temporal and seasonal conditions
- Quantizer bottleneck: 8
- Weak decoder: Num of Resnet blocks = 1, Base feature channels = 64

	Hyperparameter	rFID ↓	Codebook usage ↑	Segmentation mIoU ↑	Depth RMSE ↓
	Baseline	29.09	15.45%	34.32	6.4156
1.	Fixed masking	34.62	14.65%	35.38	5.9993
2.	Random range masking	32.07	15.67%	34.71	6.1395
3.	(2) + Data augmentation	31.60	13.94%	35.83	6.1916
4.	(3) + Diverse frames data	29.83	14.36%	36.36	6.2126
5.	(3) + Diverse image data	25.24	20.50%	33.93	6.3651
6.	(3) + Quantizer bottleneck	31.83	15.36%	34.52	6.1700
7.	(3) + Weak decoder	32.44	13.72%	34.79	6.1449

Table 12.: Results: Masked image modeling ablation under a mask ratio of 0.5. Each row represents an iterative hyperparameter modification starting from row (2) to monitor its influence on reconstruction, codebook utilization, and downstream segmentation and depth performance over VQ-VAE baseline.

Analysis :

- **Masking strategy:** Moving from fixed to random masking improves reconstruction and codebook usage, indicating better generalization, with a slight drop in segmentation.
- **Data augmentations:** Adding data augmentation improves segmentation performance but introduces a minor trade-off in depth and codebook usage.
- **Data diversity:** Incorporating diverse frame data further improves reconstruction and semantic performance, highlighting the benefit of temporal diversity. In contrast, diverse image data increases codebook usage and reconstruction quality but degrades segmentation and depth, suggesting a bias toward low-level features.
- **Architectural modifications:** Architectural changes such as a quantizer bottleneck or weak decoder increase codebook usage but demonstrate marginal improvements in semantic and geometric performance of the model.

Overall, the results indicate that a balance between diversity and moderate constraints leads to stable and generalizable representations. While these findings highlight the impact of different hyperparameters on model performance, the configuration reported in Table 2 remains the optimal setting and serves as the primary reference point for evaluation.

A.2. Impact of temporal context for auxiliary supervision using pretrained priors

From Table 4, we observe that supervision from a strong semantic pretrained teacher improves overall model performance. Building on this setting, we further explore the use of temporal context to enhance semantic representations and downstream performance. The goal is to incorporate additional temporal information to improve consistency and quality. The corresponding architecture is illustrated in Figure 26.

Pretrained Model	Masking	rFID↓	Codebook usage ↑		Segmentation mIoU ↑	Depth RMSE ↓
			CB rec	CB aux		
Baseline	✗	29.09	15.45%	NA	34.32	6.4156
Baseline Depth	✗	30.70	15.52%	42.50%	38.39	6.1160
Temporal Depth	✗	30.55	13.96%	60.16%	35.88	7.8638
Temporal Depth	✓	34.90	14.60%	57.59%	31.75	7.3607
Baseline RAFT	✗	30.87	15.53%	42.53%	36.10	6.0661
Temporal RAFT	✗	33.77	13.57%	18.02%	30.96	8.0306

Table 13.: Results: Impact of temporal context on auxiliary supervision using pretrained priors. The results demonstrate the impact of incorporating the temporal context in the setup with the supervision from pretrained priors. We notice a degraded performance under this setting in comparison to the baselines. Additionally, applying masking further deteriorates the model performance.

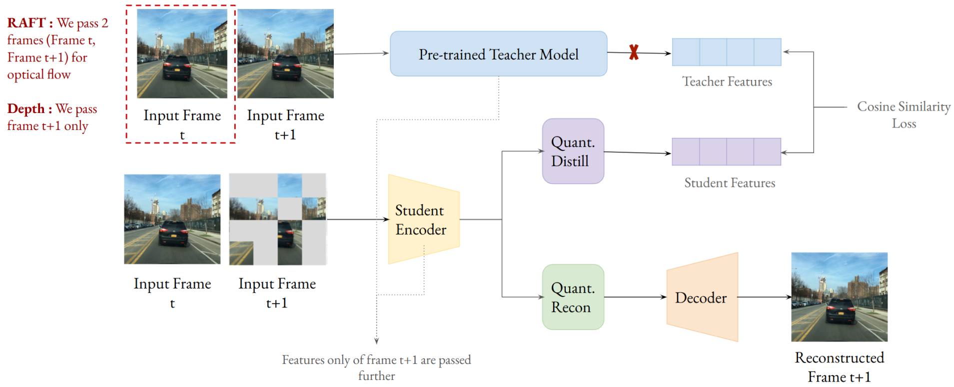


Figure 26.: Architecture: Temporal context for auxiliary supervision on student features. Temporal context is incorporated by using frame t to support supervision of the target frame t+1. The target frame is passed through an auxiliary distillation branch for supervision, while a parallel branch reconstructs it. For optical flow-based supervision (e.g., RAFT), both frames (t and t+1) are processed by the pretrained teacher, with supervision applied using features corresponding to the target frame t+1.

Here we compare the performance of the model with the VQ-VAE baseline and distillation setups with single frame input. Baseline results can be found in Table 4). From Table 13, we notice the degraded performance compared to both the standard VQ-VAE baseline and the distillation setups without temporal context (Baseline Depth and Baseline RAFT). This suggests that incorporating temporal context, with and without masking, in auxiliary supervision (distillation) setting leads to poor model performance. The masking strategy used here corresponds to the 'Single-frame Mask' setting from Input Configuration 2 (subsection 4.2.1). We do not further explore more aggressive masking configurations, as initial results already indicate consistently weak performance.

A.3. Auxiliary supervision using a pretrained MIM model

We evaluate the effects of architectural change (Figure 27) to the self-distillation experiment, where we used our own pretrained MIM model (Figure 9). The pretrained

Pretrained Model	rFID↓	Codebook usage ↑		Segmentation mIoU ↑	Depth RMSE ↓
		CB rec	CB aux		
Baseline	29.09	15.45%	NA	34.32	6.4156
Pretrained MIM (Old)	35.16	12.69%	100%	27.54	6.8400
Pretrained MIM (New)	34.54	15.19%	NA	34.11	6.0542

Table 14.: Results: Auxiliary supervision via pretrained MIM model. The teacher model is our pretrained MIM (New) model trained with a mask ratio of 0.9 via random range masking. We observe significant improvement across all the evaluation metrics in comparison to the self-distillation results (Pretrained MIM (Old)), due to the proposed architectural modifications. However the improvements still remain inferior to the VQ-VAE baseline.

MIM model is the best masking setup from subsection 4.2.1 - Input Configuration 1 with mask ratio of 90%.

The following results (Table 14) demonstrate that applying supervision in a continuous latent space improves model performance. Additionally, applying masking to the student input leads to further semantic and geometric gains compared to 'Pretrained MIM (Old)' model. However, despite these improvements, the overall performance remains inferior to the VQ-VAE baseline.

A.4. Architecture: DINO supervision for cross-frame temporal reconstruction

We evaluated the impact of DINO distillation for cross-frame temporal reconstruction (temporal compression - subsection 4.2.3) setting. For this experiment, we modified the architecture (Figure 28) by adding an extra distillation branch to the original temporal compression setting (Figure 18) for a supervision from DINO pretrained model.

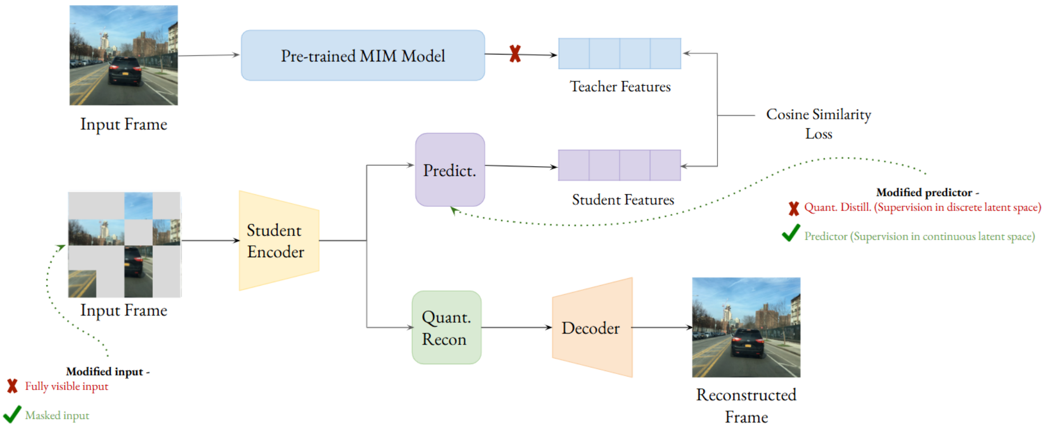


Figure 27.: Architecture: Modification of self-distillation setup under pretrained MIM model. In this setting we modify the architecture for Auxiliary supervision via Knowledge Distillation (Figure 9) to understand it effects on pretrained MIM model. Specifically, the quantizer from a distillation branch is replaced by a light weight MLP predictor, and input to the student network receives masked input.

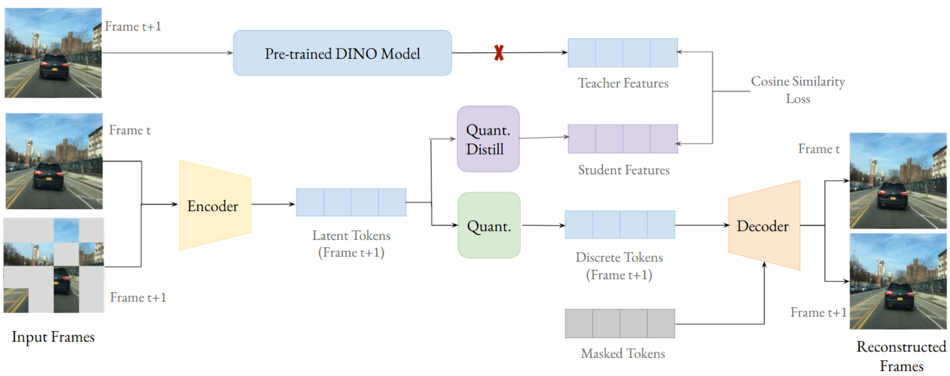


Figure 28.: Architecture : DINO distilled cross-frame temporal reconstruction. Two consecutive frames are processed jointly using a ViT encoder. Only the representation of the target masked frame (t+1) is passed through vector quantization and decoded via the reconstruction branch. In parallel, the corresponding encoded tokens are forwarded to a distillation branch, where they are aligned with teacher features using a cosine similarity loss. Mask tokens are introduced in the decoder to enable reconstruction of both frames, encouraging temporally consistent and semantically grounded representations.

The results for this setting can be found in Table 5. This setting shows improvement for temporal compression over all the evaluation metrics, while additionally showing improved depth performance for DINO distillation.

A.5. Monotonic improvement of all the metrics under EMA-stabilized teacher supervision in discrete latent space

In this setting we evaluate the performance of EMA-stabilized teacher supervision in discrete latent space (Figure 12) under varied reconstruction loss weights, with the primary results in Table 7.

Model	λ_{recon}	Masking	rFID ↓	Codebook usage ↑	Segmentation mIoU ↑	Depth RMSE ↓
Baseline	1.00	NA	29.09	15.45%	34.32	6.4156
	0.00	✗	61.55	11.72%	20.46	7.5540
	0.10	✗	45.01	13.92%	23.45	7.2550
	0.20	✗	41.54	13.94%	26.16	6.8018
	0.50	✗	37.20	14.53%	27.65	6.7517
	1.00	✗	36.74	14.31%	29.92	6.5596

Table 15.: Results: Monotonic improvement under EMA supervision in discrete latent space. We observe a generally monotonic improvement across all the evaluation metrics under increased reconstruction loss weight setting, when supervision is applied in the discrete latent space. However, this design choice shows limited performance required to improve over the baseline VQ-VAE model.

For EMA-stabilized supervision in the discrete latent space, we observe a generally monotonic improvement across all the evaluation metrics under increased reconstruction loss weight with cosine loss weight fixed at 1.0 ($\lambda_{cosine} = 1.0$). Table 15 demonstrates the trend showing the stabilizing force of reconstruction loss in 'EMA-stabilized teacher supervision in discrete latent space' setting.

A.6. Architecture: Decoupled quantization setting for EMA-stabilized teacher supervision in discrete latent space

During EMA-based supervision in the discrete token space (Figure 12), we analyze the impact of introducing additional flexibility by separating the quantizer into a separate auxiliary branch (Quant aux). Although this architectural modification (Figure 29) significantly increases codebook utilization in the auxiliary branch, it does not lead to improvements across the evaluation metrics.

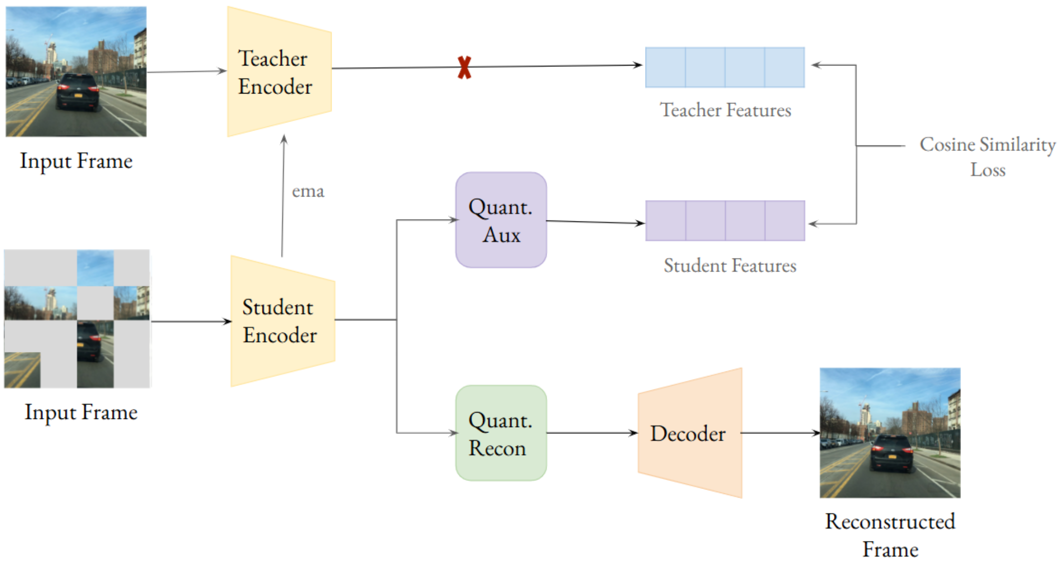


Figure 29.: Architecture: EMA-supervision in the discrete token space under decoupled quantizer setting. In this setting an EMA-supervision is applied in the discrete token space. This architecture builds upon Figure 12 by introducing a decoupled design with a separate auxiliary quantizer branch, providing additional flexibility in learning representations.

A.7. Visualizations: Sampled images

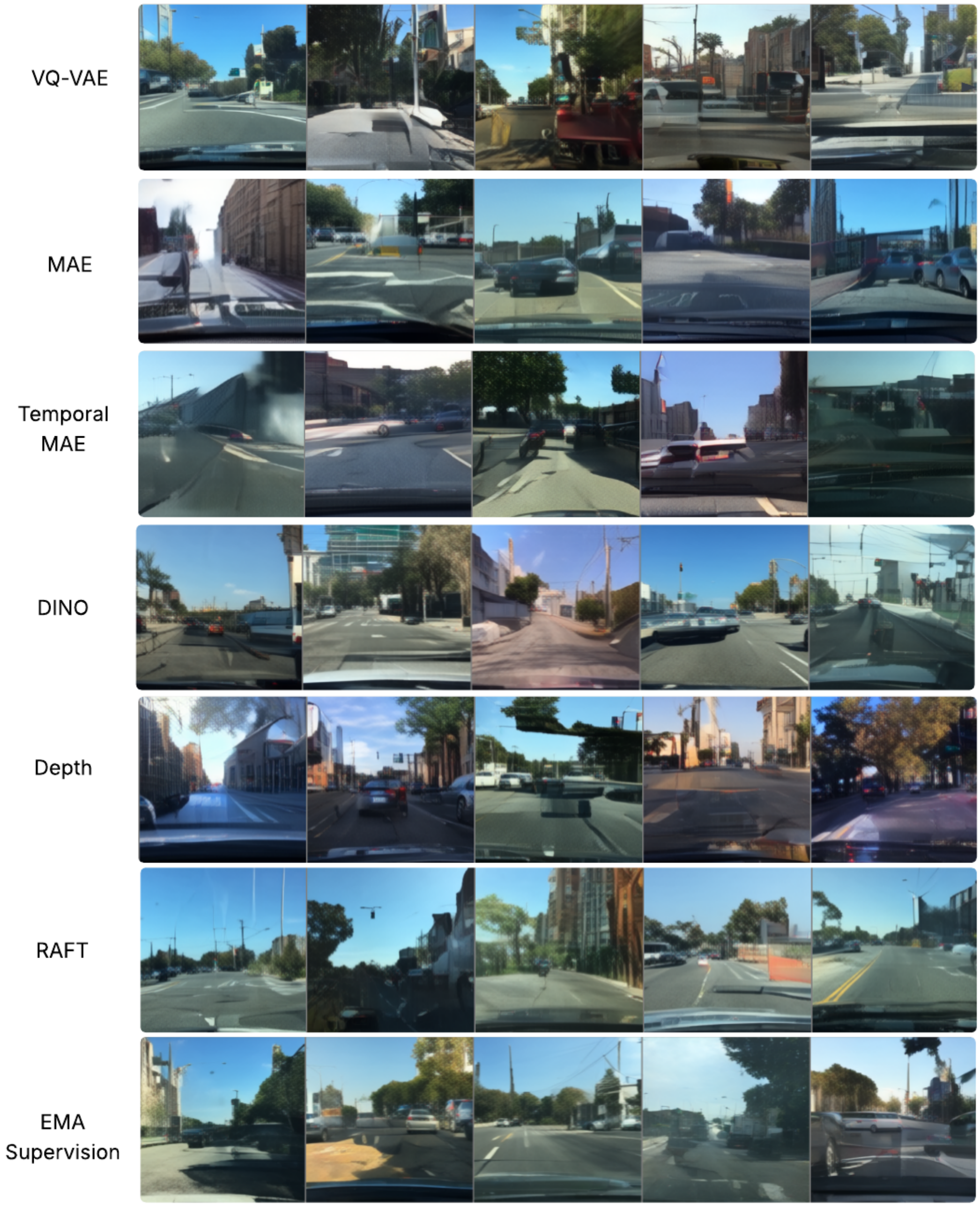


Figure 30.: Sampled Images for the best tokenizer configurations. Qualitative samples of generated images from the best-performing tokenizer configurations, highlighting differences in perceptual quality and structural consistency.

A.8. Use of Large Language Models (LLMs)

We use ChatGPT (OpenAI) as a coding assistant during implementation and for language refinement. All core contributions and the initial draft are produced by the student.

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